



Analog IC Design Implementation and Verification
Weeks 7 and 8 – Advanced Amplifier Topologies and Design

Version IC 23.1 ISR11.29 with SPECTRE 24.1 ISR1

Revision 1.0

Estimated time: 1 Week

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Introduction to Opamp Topologies

You have several degrees of freedom and choices of different circuit architectures that can enhance the performance of a given opamp:

- Buffered opamps
- High-speed/high-frequency opamps
- Low-power opamps
- Low-voltage opamps

Note: Some of these techniques have already been described. However, these techniques are shown here in relation to the design of an opamp, not just a single-amplifier stage.



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What Is a Buffered Opamp?

- A buffered opamp is an opamp with a low value of output resistance, R_o , and typically is:

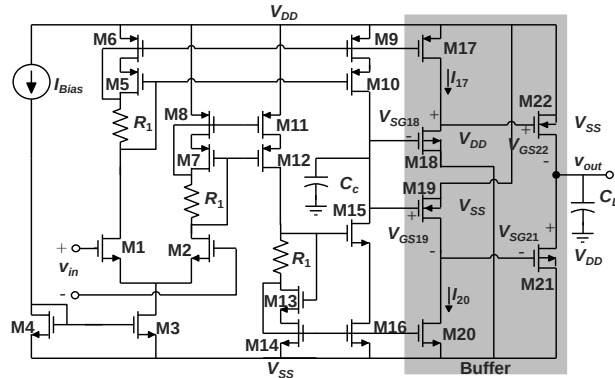
$$10\Omega \leq R_o \leq 1000\Omega$$

- Requirements are generally the same as for the output amplifier:
 - Low-output resistance
 - Large output-signal swing
 - Low distortion
 - High efficiency
- Types of buffered opamps
 - Buffered opamps using MOSFETs with and without negative feedback
 - Buffered opamps using BJTs



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Source-Follower Push-Pull-Output Opamp



$$R_{out} = \frac{1}{g_{m21} + g_{m22}} \leq 1000\Omega, A_v(0) = 65dB(I_{Bias} = 50\mu A), \text{ and } GB = 60MHz \text{ for } C_L = 1pF$$

What is the output bias current?

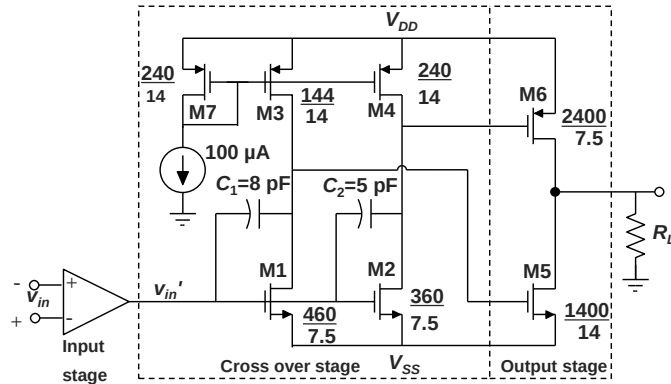
$$M18 - M19 - M21 - M22 \text{ loop} \Rightarrow V_{SG18} + V_{GS19} = V_{SG21} + V_{GS22}$$

$$\text{This gives } \sqrt{\frac{2I_{18}}{K_P S_{18}}} + \sqrt{\frac{2I_{19}}{K_N S_{19}}} = \sqrt{\frac{2I_{21}}{K_P S_{21}}} + \sqrt{\frac{2I_{22}}{K_N S_{22}}}$$

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Crossover-Inverter Buffer-Stage Opamp

- **Principle:** Suppose the buffer has high output resistance and voltage gain (common source). This is okay if the gain is approximately unity, when the stage is loaded by a small R_L .

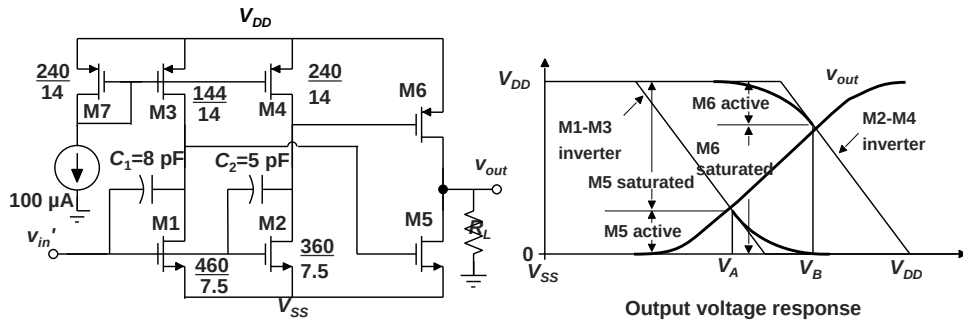


- This opamp can deliver 160 mW to a 100Ω load while dissipating only 7 mW of quiescent power!

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How Does the Output Buffer Work?

- The two inverters, M1-M3 and M2-M4 are designed to work over different regions of the buffer input voltage, v_{in}' . Consider the idealized voltage-transfer characteristic of the crossover inverters:



- Crossover voltage $\equiv V_C = V_B - V_A \geq 0$
- V_C is designed to be small and positive for worst-case variations in processing (maximum value of $V_C \approx 110\ \text{mV}$).

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Performance Results

The table shows the performance results for the crossover-inverter, buffer stage CMOS opamp.

Specification	Performance
Supply voltage	$\pm 6\text{V}$
Quiescent power	7 mW
Output swing (100 Ω Load)	8.1 V _{pp}
Open-loop gain (100 Ω Load)	78.1 dB
Unity-gain bandwidth	260 kHz
Voltage spectral noise density at 1 kHz	1.7 $\mu\text{V}/\sqrt{\text{Hz}}$
PSRR at 1 kHz	55 dB
CMRR at 1 kHz	42 dB
Input offset voltage (typical)	10 mV



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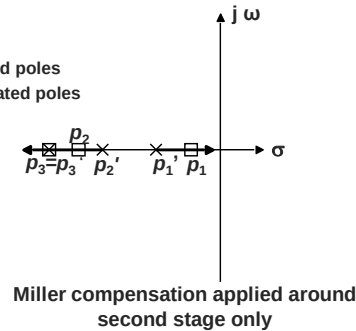
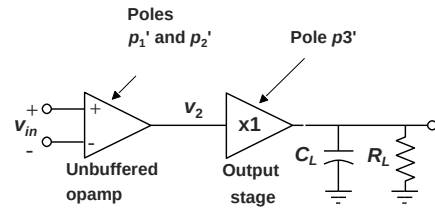
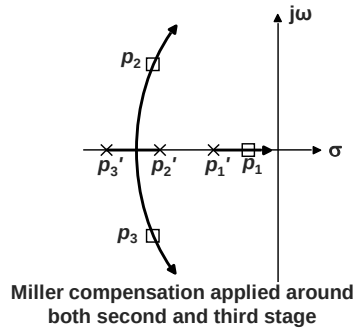
Compensation of Opamps with Output Amplifiers

Compensation of a three-stage amplifier:

- The output opamp introduces a third pole, p'_3 .
- With no compensation:

$$\frac{V_{out}(s)}{V_{in}(s)} = \frac{-A_{vo}}{\left(\frac{s}{p'_1} - 1\right)\left(\frac{s}{p'_2} - 1\right)\left(\frac{s}{p'_3} - 1\right)}$$

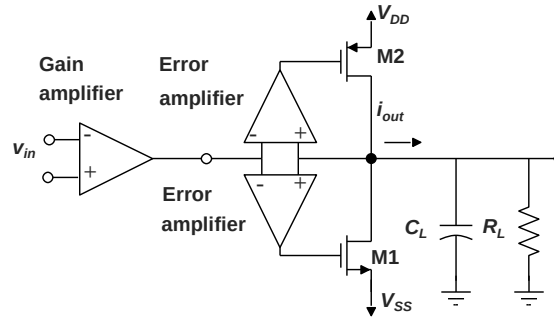
- The compensation choices are:



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Low Output-Resistance Opamp

To get low output resistance with MOSFETs, negative feedback must be used.



Ideal implementation of low output-resistance opamp

- The output resistance is equal to $r_{ds1} || r_{ds2}$ divided by the loop gain.
- If the error amplifiers are not perfectly matched, the bias current in M1 and M2 is not defined.

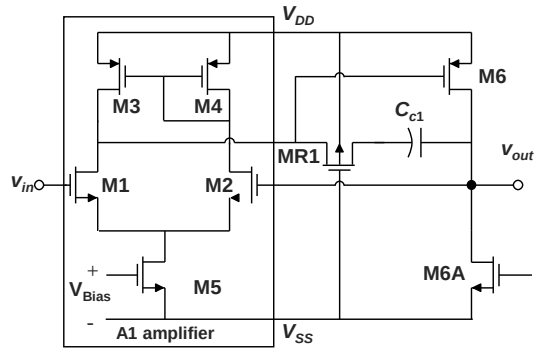
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The circuit works the following way:

- A similar result holds for negative offsets and offsets in E_{A2} .



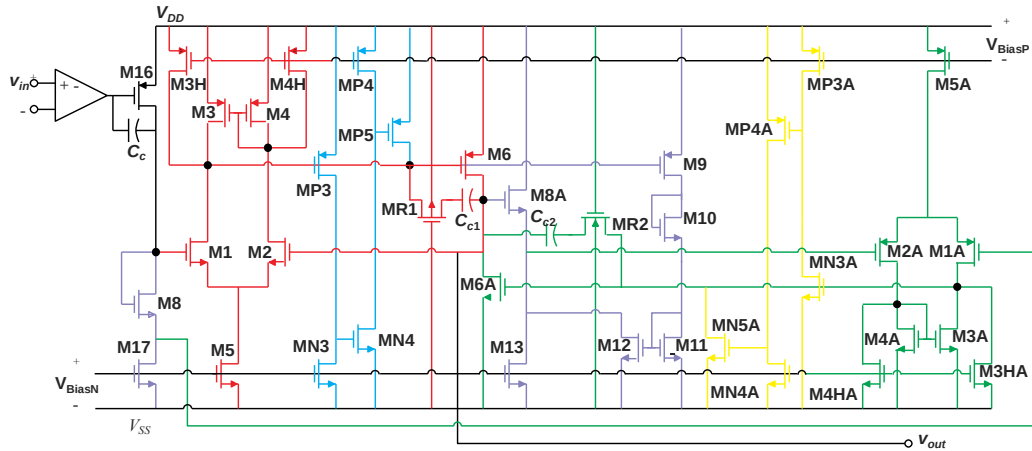
Low Output-Resistance Opamp (continued)



Error amplifier circuit

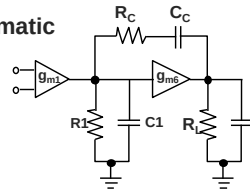
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Low Output-Resistance Opamp (continued)



Complete low-output resistance opamp schematic

- Compensation: Uses nulling Miller compensation
- Short circuit protection:
- MP3-MN3-MN4-MP4-MP5
- MN3A-MP3A-MP4A-MN4A-MN5A (max. output ± 60 mA)



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Low Output Resistance Opamp (continued)

Performance characteristics of the low output-resistance opamp		
Specification	Simulated results	Measured results
Power dissipation	7.0 mW	5.0 mW
Open loop voltage gain	82 dB	83 dB
Unity-gain bandwidth	500 kHz	420 kHz
Input offset voltage	0.4 mV	1 mV
PSRR*(0)/PSRR*(0)	85 dB/104 dB	86 dB/106 dB
PSRR*(1 kHz)/PSRR*(1 kHz)	81 dB/98 dB	80 dB/98 dB
THD ($V_{in} = 3.3 V_{pp}$) $R_L = 300\Omega$ $C_L = 1000 \text{ pF}$	0.03% 0.08%	0.13%(1 kHz) 0.32%(4 kHz)
THD ($V_{in} = 4.0 V_{pp}$) $R_L = 15 \text{ K}\Omega$ $C_L = 200 \text{ pF}$	0.05% 0.16%	0.13%(1 kHz) 0.20%(4 kHz)
Settling time (0.1%)	3 μs	<5 μs
Slew rate	0.8 V/ μs	0.6 V/ μs
1/f Noise at 1 kHz	-	130 nV/ $\sqrt{\text{Hz}}$
Broadband noise	-	49 nV/ $\sqrt{\text{Hz}}$

$$R_{out} \approx \frac{r_{ds6} \parallel r_{ds6A}}{\text{Loop gain}} \approx \frac{50\text{k}\Omega}{5000} = 10\Omega$$



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Low Output-Resistance Opamp (continued)

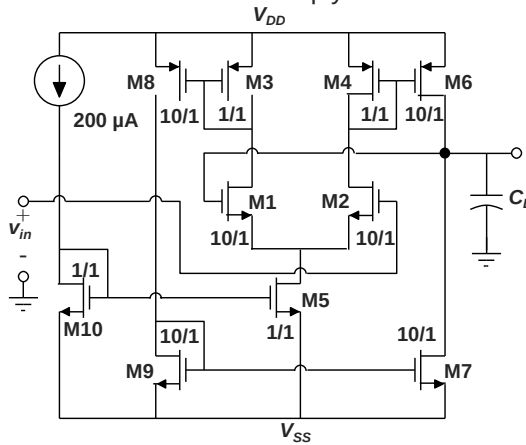
Component sizes for the low-resistance opamp			
Transistor/ capacitor	$\mu\text{m}/\mu\text{m}$ or pF	Transistor/ capacitor	$\mu\text{m}/\mu\text{m}$ or pF
M16	184/9	M8A	481/6
M17	66/12	M13	66/12
M8	184/6	M9	27/6
M1, M2	36/10	M10	6/22
M3, M4	194/6	M11	14/6
M3H, M4H	16/12	M12	140/6
M5	145/12	MP3	8/6
M6	2647/6	MN3	244/6
MRC	48/10	MP4	43/12
C_C	11.0	MN4	12/6
M1A, M2A	88/12	MP5	6/6
M3A, M4A	196/6	MN3A	6/6
M3HA, M4HA	10/12	MP3A	337/6
M5A	229/12	MN4A	24/12
M6A	2420/6	MP4A	20/12
C_F	10.0	MN5A	6/6



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Simpler Implementation of Negative Feedback

- Negative feedback can be done more simply to achieve low output resistance.



Output resistance is:

$$R_{out} = \frac{R_o}{1 + LG}$$

where:

$$R_o = \frac{1}{g_{ds6} + g_{ds7}}$$

and:

$$|LG| = \frac{g_{m2}}{2g_{m4}} (g_{m6} + g_{m7}) R_o$$

- Therefore, the output resistance is given by this formula:

$$R_{out} = \frac{1}{(g_{ds6} + g_{ds7}) \left[1 + \frac{g_{m2}}{2g_{m4}} (g_{m6} + g_{m7}) R_o \right]}$$

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Using the Shunt Negative-Feedback Buffer

- Find the output resistance of previous opamp using the following model parameters:
 - $C_{ox} = 24.7 \times 10^{-4} \text{ F/m}^2$
 - $K_n' = 110 \times 10^{-6} \text{ A/V}^2$ $K_p' = 50 \times 10^{-6} \text{ A/V}^2$
 - $KF = 4 \times 10^{-28} \text{ F} \cdot \text{A}$ for NMOS, $KF = 0.5 \times 10^{-28} \text{ F} \cdot \text{A}$ for PMOS
 - $\lambda_N = 0.03 \text{ V}^{-1}$ $\lambda_P = 0.06 \text{ V}^{-1}$

Solution

- The current flowing in the output transistors, M6 and M7, is 1 mA, which gives R_o of:

$$R_o = \frac{1}{(\lambda_N + \lambda_P) 1 \text{ mA}} = \frac{1000}{0.09} = 11.11 \text{ k}\Omega$$

- To calculate the loop gain, you find that:

$$g_{m2} = \sqrt{2K_N' \cdot 10 \cdot 100 \mu\text{A}} = 469 \mu\text{S}$$

and

$$g_{m4} = \sqrt{2K_P' \cdot 1 \cdot 100 \mu\text{A}} = 100 \mu\text{S}$$

$$g_{m6} = \sqrt{2K_P' \cdot 10 \cdot 1000 \mu\text{A}} = 1 \mu\text{S}$$

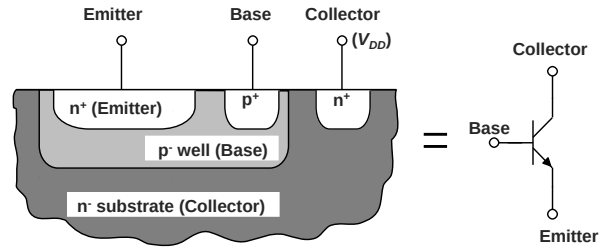
- Therefore, the loop gain is: $|LG| = \frac{469}{100} 12 \cdot 11.11 = 104.2$
- Solving for the output resistance, R_{out} , gives: $R_{out} = \frac{11.11 \text{ k}\Omega}{1 + 104.2} = 106 \Omega$



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BJTs Available in CMOS Technology

The figure shows an NPN substrate BJT available in a p-well CMOS technology.



- g_m of the BJT is larger than the FET so that the output resistance without feedback is lower.
- You can use the lateral or substrate BJT but because the collector is on AC ground, the substrate BJT is preferred.
- Current is required to drive the BJT.

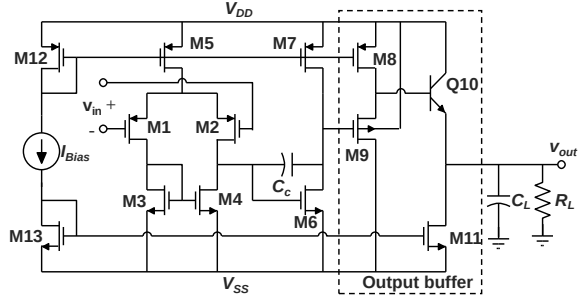
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Two-Stage Opamp with a Class-A BJT Output Buffer Stage

- The purpose of the M8-M9 source follower is to:
 - Reduce the output resistance (includes whatever is seen from the base to ground divided by $1+\beta_F$).
 - Reduce the output load at the drains of M6 and M7.

- Small-signal output resistance is:

$$R_{out} \approx \frac{r_{\pi 10} + (1/g_{m9})}{1 + \beta_f} = \frac{1}{g_{m10}} + \frac{1}{g_{m9}(1 + \beta_F)}$$



Two-stage opamp with a class-A BJT output buffer stage

- Maximum output voltage is: $= 51.6\Omega + 6.7\Omega = 58.3\Omega$ where $I_{10} = 500\mu A$, $I_8 = 100\mu A$, $W_9/L_9 = 100$ and β_F is 100

$$v_{OUT}(\max) = V_{DD} - V_{SD8}(\text{sat}) - v_{BE10} = V_{DD} - \sqrt{\frac{2K_p'}{I_8(W_8/L_8)}} - V_t \ln\left(\frac{I_{c10}}{I_{s10}}\right)$$

- Voltage gain is:

$$\frac{v_{out}}{v_{in}} = \left(\frac{g_{m1}}{g_{ds2} + g_{ds4}} \right) \left(\frac{g_{m6}}{g_{ds6} + g_{ds7}} \right) \left(\frac{g_{m9}}{g_{m9} + g_{mbs9} + g_{ds8} + g_{\pi 10}} \right) \left(\frac{g_{m10} R_L}{1 + g_{m10} R_L} \right)$$

Compensation is more complex because of the additional stages.



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Designing Class A Buffered Opamps

Using the parameters of shunt negative-feedback buffer discussed previously along with the BJT parameters of $I_S = 10\text{-}14\text{A}$ and $\beta_F = 100$, design the class A buffered opamp to give the following specifications. Assume the channel length is to be $1\text{ }\mu\text{m}$.

$V_{DD} = 2.5\text{V}$	$V_{SS} = -2.5\text{V}$	$GB = 5\text{ MHz}$	$A_{vd}(0) \geq 5000\text{ V/V}$
$CL = 100\text{ pF}$	Slew rate $\geq 10\text{ V}/\mu\text{s}$	$RL = 500\Omega$	$R_{out} \leq 100\Omega$
$ICMR = -1\text{V to } 2\text{V}$			

Solution

Because the previous specifications are similar to the two-stage design of the shunt negative-feedback buffer, you can use these results for the first two stages of the design. However, you must convert the results of shunt negative-feedback buffer to a PMOS input stage. The results of doing this give $W_1 = W_2 = 6\text{ }\mu\text{m}$, $W_3 = W_4 = 7\text{ }\mu\text{m}$, $W_5 = 11\text{ }\mu\text{m}$, $W_6 = 43\text{ }\mu\text{m}$, and $W_7 = 34\text{ }\mu\text{m}$.

BJT follower: $SR = 10\text{V}/\mu\text{s}$ and 100pF capacitor give $I_{11} = 1\text{mA}$

$$\therefore \text{ If } W_{13} = 44\mu\text{m}, \text{ then } W_{11} = 44\mu\text{m}(1000\mu\text{A} / 30\mu\text{A}) = 1467\mu\text{m}$$

$$I_{11} = 1\text{mA} \Rightarrow 1/g_{m10} = 0.0258\text{V}/\text{mA} = 25.8\Omega$$

MOS follower: To source 1 mA , the BJT must provide 2 mA which requires $20\text{ }\mu\text{A}$ from the MOS follower. Therefore, select a bias current of $100\text{ }\mu\text{A}$ for M8.

If $W_{12} = 44\text{ }\mu\text{m}$, then $W_8 = 44\text{ }\mu\text{m} (100\text{ }\mu\text{A}/30\text{ }\mu\text{A}) = 146\text{ }\mu\text{m}$.



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Designing Class-A Buffered Opamps (continued)

- If $1/g_{m10}$ is 25.8W, then design g_{m9} as:

$$g_{m9} = \frac{1}{[R_{out} - (1/g_{m10})](1 + \beta_F)} = \frac{1}{[100 - 25.8](101)} = 133.4 \mu S \quad g_{m9} \text{ and } I_9 \Rightarrow W/L = 0.809$$

Select $W/L = 10$ for M9 to make sure that the contribution of M9 to the output resistance is sufficiently small and to increase the gain closer to unity. This gives a transconductance of M9 of $469 \mu S$.

- To calculate the voltage gain of the MOS follower you need to find g_{mbs9} :

$$\therefore g_{mbs9} = \frac{g_{m9} \gamma_N}{2\sqrt{2\phi_F + V_{BS9}}} = \frac{469 \cdot 0.4}{2\sqrt{0.7 + 2}} = 57.1 \mu S$$

where the value of V_{BS9} is assumed to be approximately 2V.

$$\therefore A_{MOS} = \frac{469 \mu S}{469 \mu S + 57.1 \mu S + 4 \mu S + 5 \mu S} = 0.8765 V/V$$

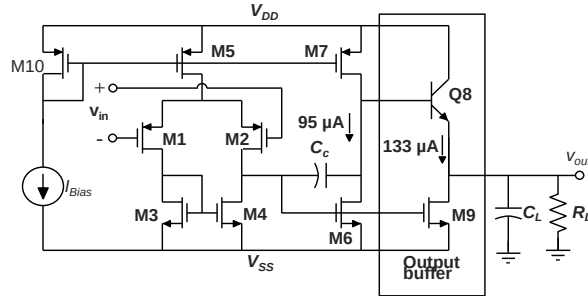
- The voltage gain of the BJT follower is: $A_{BJT} = \frac{500}{25.8 + 500} = 0.951 V/V$
- Thus, the gain of the opamp is: $A_{vd}(0) = (7777)(0.8765)(0.951) = 6483 V/V$
- The power dissipation of this amplifier is: $P_{diss} = 5V(1255 \mu A) = 6.27 mW$



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Two-Stage Opamp: Class AB BJT Output Buffer Stage

This amplifier can reduce the quiescent power dissipation.

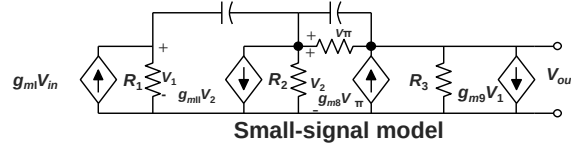


- The slew rate is: $SR^+ = \frac{I_{OUT}^+}{C_L} = \frac{(1 + \beta_F)I_7}{C_L}$ and $SR^- = \frac{\beta_9(V_{DD} - 1V + |V_{SS}| - V_{T0})^2}{2C_L}$
- If $\beta_F = 100$, $C_L = 1000$ pF and $I_7 = 95$ μ A, then $SR^+ = 8.59$ V/ μ s.
- Assuming a $W_9/L_9 = 60$ ($I_9 = 133$ μ A), ± 2.5 V power supplies and $C_L = 1000$ pF gives $SR^- = 35.9$ V/ μ s.
- The current is not limited by I_7 as it is for the positive slew rate.

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BJT Output Buffer Stage (continued)

- Small-signal characteristics are:



- The nodal equations are:

$$\begin{aligned} g_{mI}V_{in} &= (G_I + sC_c)V_1 - sC_cV_2 + 0V_{out} \\ 0 &= (g_{mII} - sC_c)V_1 + (G_{II} + g_{\pi} + sC_c + sC_{\pi})V_2 - (g_{\pi} + sC_{\pi})V_{out} \\ 0 &\cong g_{m9}V_1 - (g_{m13} + sC_{\pi})V_2 + (g_{m13} + sC_{\pi})V_{out} \quad \text{where } g_{\pi} > G_3 \end{aligned}$$

- The approximate voltage-transfer function is:

$$\frac{V_{out}(s)}{V_{in}(s)} \approx A_{v0} \frac{((s/z_1) - 1)((s/z_2) - 1)}{((s/p_1) - 1)((s/p_2) - 1)}$$

Where:

$$\begin{aligned} A_{v0} &= \frac{-g_{mI}g_{mII}}{G_I G_{II}} & z_1 &= \frac{1}{\frac{C_c}{g_{mII}} - \frac{C_{\pi}}{g_{m13}} \left[1 + \frac{g_{m9}}{g_{mII}} \right]} & z_2 &= -\frac{g_{m13}}{C_{\pi}} + \frac{g_{mII}}{C_c} \left[1 + \frac{g_{m9}}{g_{mII}} \right] \\ p_1 &= \frac{-G_I G_{II}}{g_{mII} C_c} \left[1 + \frac{g_{m9}}{\beta_F g_{mII}} + \frac{C_{\pi}}{C_c} \left(\frac{G_I G_{II}}{g_{m13} g_{mII}} \right) \right]^{-1} & p_2 &\cong \frac{-g_{m13} g_{mII}}{(g_{mII} + g_{m9}) C_{\pi}} \end{aligned}$$



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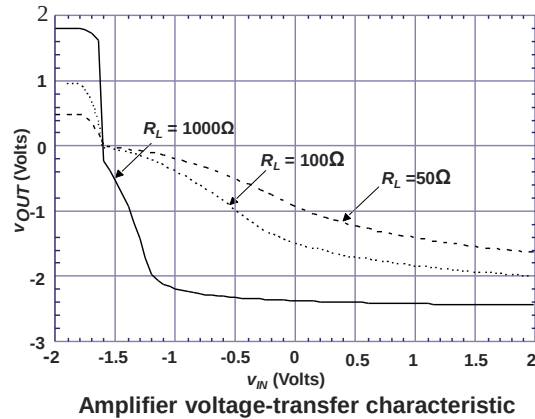
BJT Output Buffer Stage (continued)

- Output stage current, I_{C8} , is: $I_{C8} = I_{D9} = \frac{S_9}{S_6} I_{D6} = \frac{60}{43} 95 \mu A = 133 \mu A$

$$r_{out} = \frac{r_{\pi} + R_{II}}{1 + \beta_F} = \frac{19.668 k\Omega + 116.96 k\Omega}{101} = 1353 \Omega$$

- Small-signal output resistance is if $I_6 = I_7 = 95 \mu A$, and $\beta_F = 100$.

The loading effect of R_L on the voltage-transfer curve is shown in the figure. (Increasing W_9/L_9 improves the negative part at the cost of power dissipation.)



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Performance: Two-Stage Class AB Output Buffer

Using the transistor currents given previously for the output stages (output stage of the two-stage opamp and the buffer stage), find the small-signal output resistance and the maximum output voltage when $R_L = 50\Omega$. Assume that the NPN BJT has the parameters of $\beta_F = 100$ and $I_S = 10 \text{ fA}$.

Solution

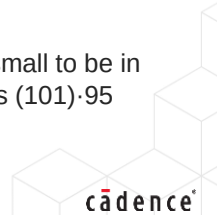
- It was shown on the previous slide that the small-signal output resistance is:

$$r_{out} = \frac{r_{\pi} + r_{ds6} \parallel r_{ds7}}{1 + \beta_F} = \frac{19.668k\Omega + 116.96k\Omega}{101} = 1353\Omega$$

- This maximum output voltage is valid only if the load current is small. If this is not the case, then a better approach is to assume that all of the current in M7 becomes base current for Q8. This base current is multiplied by $1 + \beta_F$ to give the sourcing current. If M9 is off, then all this current flows through the load resistor to give an output voltage of:

$$v_{OUT}(\max) \approx (1 + \beta_F) I_7 R_L$$

- If the value of $v_{OUT(\max)}$ is close to V_{DD} , then the source-drain voltage across M7 might be too small to be in saturation, causing I_7 to decrease. Using the previous equation, you can calculate $v_{OUT(\max)}$ as $(101) \cdot 95 \mu\text{A} \cdot 50\Omega$ or 0.48V , which is close to the simulation results shown previously.



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Summary: Buffered Opamps

A buffered opamp requires an output resistance that is between the values $10\Omega \leq R_o \leq 1000\Omega$:

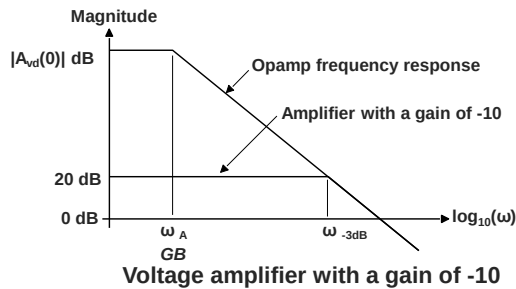
- Output resistance using MOSFETs can be reduced only by:
 - Source follower output ($1/g_m$)
 - Negative shunt feedback (frequency is a problem in this approach)
- Use of substrate (or lateral) BJTs can reduce the output resistance because g_m is larger than the g_m of a MOSFET.
- Adding a buffer stage to lower the output resistance will most likely complicate the compensation of the opamp.



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What Is the Influence of GB on Frequency Response?

The opamp is designed primarily to be used with negative feedback. When the product of the opamp gain and feedback gain (loss) is not greater than unity, negative feedback does not work satisfactorily.



What causes the GB?

In fact, $GB = \frac{g_m}{C}$

where g_m is the transconductance that converts the input voltage to current and C is the capacitor that causes the dominant pole. This relationship assumes that all higher-order poles are greater than GB.

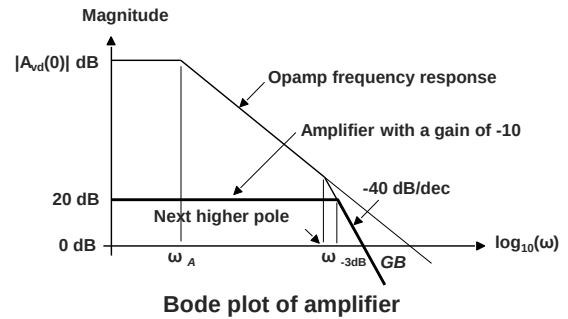
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What Is the Limit of GB?

The figure shows what happens when the next higher pole is *not* greater than GB.

For a two-stage opamp, the poles and zeros are the following:

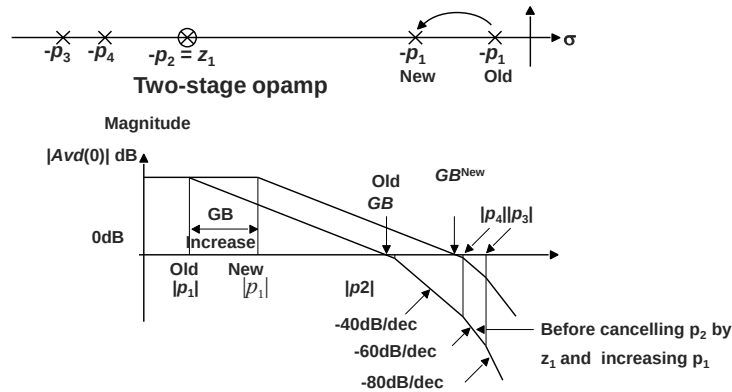
- Dominant pole $p_1 = \frac{-g_{m1}}{A_v(0)C_c}$
- Output pole $p_2 = \frac{-g_{m6}}{C_L}$
- Mirror pole $p_3 = \frac{-g_{m3}}{C_{gs3} + C_{gs4}}$
- Nulling pole $p_4 = \frac{-1}{R_z C_I}$
- Nulling zero $z_1 = \frac{-1}{R_z C_c - (C_c / g_{m6})}$



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Procedure: Increasing GB of Two-Stage Opamps

1. Use the nulling zero to cancel the closest pole beyond the dominant pole.
2. Set the maximum GB equal to the magnitude of the second closest pole beyond the dominant pole.
3. Adjust the dominant pole so that $GB \gg 2.2 \times (\text{second closest pole beyond the dominant pole})$. The figure shows a two-stage opamp where p_2 is the next closest pole beyond the dominant pole.



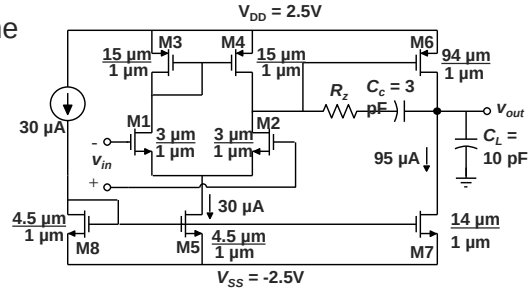
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Example: Increasing GB of Opamps

Use the two-stage opamp at right and apply the procedure to increase the gain-bandwidth as much as possible.



Solution

1. First find the values of p_2 , p_3 , and p_4 .

- It can be shown that $p_2 = -94.25 \times 10^6 \text{ rads/sec}$ and $p_3 = -2.81 \times 10^6 \text{ rads/sec}$
- To find p_4 , you must find C_1 which is the output capacitance of the first stage of the opamp. C_1 consists of the following capacitors:

$$C_1 = C_{bd2} + C_{bd4} + C_{gs6} + C_{gd2} + C_{gd4}$$

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Example: Increasing GB of Opamps (continued)

For C_{bd2} the width is $3\text{ }\mu\text{m} \rightarrow L1+L2+L3 = 3\text{ }\mu\text{m} \rightarrow AS/AD=9\text{ }\mu\text{m}^2$ and $PS/PD = 12\text{ }\mu\text{m}$.

For C_{bd4} the width is $15\text{ }\mu\text{m} \rightarrow L1+L2+L3 = 3\text{ }\mu\text{m} \rightarrow AS/AD=45\text{ }\mu\text{m}^2$ and $PS/PD = 36\text{ }\mu\text{m}$.

From a table:

$$C_{bd2} = (9\text{ }\mu\text{m}^2)(770 \times 10^{-6}\text{ F/m}^2) + (12\text{ }\mu\text{m})(380 \times 10^{-12}\text{ F/m}) = 6.93\text{ fF} + 4.56\text{ fF} = 11.5\text{ fF}$$

$$C_{bd4} = (45\text{ }\mu\text{m}^2)(560 \times 10^{-6}\text{ F/m}^2) + (36\text{ }\mu\text{m})(350 \times 10^{-12}\text{ F/m}) = 25.2\text{ fF} + 12.6\text{ fF} = 37.8\text{ fF}$$

C_{gs6} is given by the following equation:

$$\begin{aligned} C_{gs6} &= CGDO \times W_6 + 0.67(C_{ox}W_6L_6) = (220 \times 10^{-12})(94 \times 10^{-6}) + (0.67)(24.7 \times 10^{-4})(94 \times 10^{-12}) \\ &= 20\text{ fF} + 154.8\text{ fF} = 175.5\text{ fF} \end{aligned}$$

$$C_{gd2} = 220 \times 10^{-12} \times 3\text{ }\mu\text{m} = 0.66\text{ fF} \text{ and } C_{gd4} = 220 \times 10^{-12} \times 15\text{ }\mu\text{m} = 3.3\text{ fF}$$

Therefore: $C_l = 11.5\text{ fF} + 37.8\text{ fF} + 175.5\text{ fF} + 0.66\text{ fF} + 3.3\text{ fF} = 228.8\text{ fF}$

Although C_{bd2} and C_{bd4} is reduced with a reverse bias, use these values to provide a margin. In fact, you should double the whole capacitance to make sure that other layout parasitics are included. Thus let C_l be 300 fF.



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Example: Increasing GB of Opamps (continued)

R_z can be calculated as 4.591 k Ω which gives $p_4 = -0.726 \times 10^9$ rads/sec.

- Using the nulling zero, z_1 , to cancel p_2 , gives p_4 as the next smallest pole.
- For 60° phase margin, $GB = |p_4|/2.2$ if the next smallest pole is more than 10GB.

$$\therefore GB = 0.726 \times 10^9 / 2.2 = 0.330 \text{ rads/sec or } 52.5 \text{ MHz}$$

- This value of GB is designed from the relationship that $GB = g_{m1}/C_c$.
- Assuming g_{m1} is constant, then:

$$C_c = g_{m1} / GB = (94.25 \times 10^{-6}) / (0.330 \times 10^9) = 286 \text{ fF}$$

- It might be useful to increase g_{m1} to keep C_c above the surrounding parasitic capacitors ($C_{gd6} = 20.7$ fF).
- The success of this method assumes that there are no other roots with a magnitude smaller than 10GB.

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Conclusion: Increasing GB of Opamps

Maximum GB depends on the input transconductance and the capacitance that causes the dominant pole.

Quantity	MOSFET opamp	BJT opamp
g_m dependence	$\sqrt{2K' \frac{W}{L} I_D}$	$\frac{I_C}{kT/q} = \frac{I_C}{V_i}$
Maximum g_m	$\approx 1 \text{ mA/V}$	$\approx 20 \text{ mA/V}$
GB for 10 pF	15 MHz	300 MHz
GB for 1 pF	150 MHz	3 GHz

The power dissipation is large for large GB because current is needed for large g_m .

Assumptions

- All higher-order roots are higher than GB.
- The larger the GB, the more difficult this becomes.

Conclusions

- The best CMOS opamps have a GB of 10-50 MHz.
- The best BJT opamps have a GB of 100-200 MHz.

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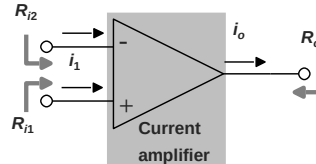
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Current Feedback Opamps

Why current feedback?

- Higher GB
- Less voltage swing → more dynamic range

What is a current amplifier?



Requirements

$$i_o = A_i(i_1 - i_2)$$

$$R_{i1} = R_{i2} = 0\Omega$$

$$R_o = \infty$$

Ideal source and load requirements

$$R_{source} = \infty$$

$$R_{Load} = 0\Omega$$

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Current Feedback Amplifier: Bandwidth Advantage

Consider the inverting voltage amplifier shown using a current amplifier with negative current feedback:

- The output current, i_o , of the current amplifier can be written as:

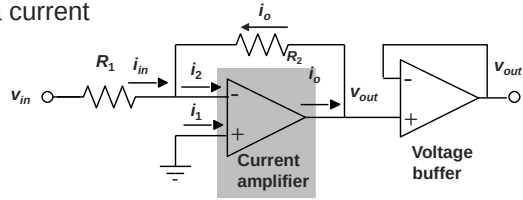
- The closed-loop current gain, i_o/i_{in} , can be found as:

$$i_o = A_i(s)(i_1 - i_2) = -A(s)(i_{in} - i_o)$$

- However, $v_{out} = i_o R_2$ and $v_{in} = i_{in} R_1$. Solving for the voltage gain, v_{out}/v_{in} gives $\frac{i_o}{i_{in}} = \frac{-A_i(s)}{1 + A_i(s)}$

$$\frac{v_{out}}{v_{in}} = \frac{i_o R_2}{i_{in} R_1} = \left(\frac{-R_2}{R_1} \right) \left(\frac{-A_i(s)}{1 + A_i(s)} \right)$$

$$\text{If } A_i(s) = \frac{A_o}{\frac{s}{\omega_A} + 1}, \text{ then } \frac{v_{out}}{v_{in}} = \left(\frac{-R_2}{R_1} \right) \left(\frac{A_o}{1 + A_o} \right) \left(\frac{\omega_A(1 + A_o)}{s + \omega_A(1 + A_o)} \right) \Rightarrow A_v(0) = \frac{-R_2 A_o}{R_1(1 + A_o)} \text{ and } \omega_{-3dB} = \omega_A(1 + A_o)$$



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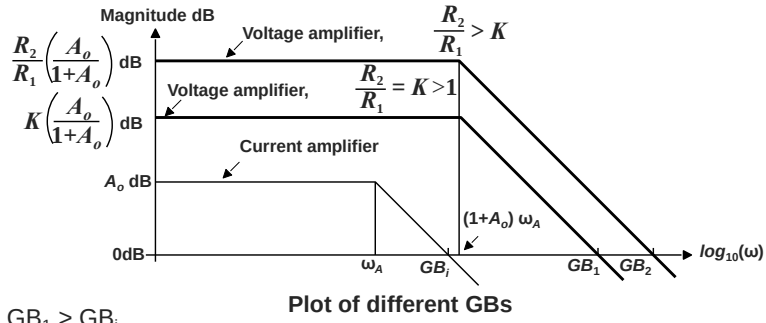
Bandwidth Advantage (continued)

- The unity-gain bandwidth (GB) is:

$$GB = |A_v(0)|\omega_{-3dB} = \frac{R_2 A_o}{R_1 (1 + A_o)} \cdot \omega_A (1 + A_o) = \frac{R_2}{R_1} A_o \cdot \omega_A = \frac{R_2}{R_1} GB_i$$

where GB_i is the unity-gain bandwidth of the current amplifier.

- If GB_i is constant, then increasing R_2/R_1 (the voltage gain) increases GB.

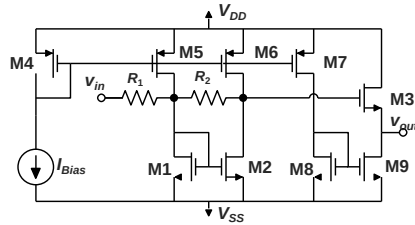


- Notice that $GB_2 > GB_1 > GB_i$.
- In the figure, it is assumed that the GB of the voltage amplifier realizing the voltage buffer is greater than the GB achieved from the previous method.

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High-Frequency Amplifier: Simple Current Mirror

- Because the gain of the current amplifier does not need to be large, consider a unity-gain current mirror implementation:



- An inverting amplifier with a gain of 10 is achieved if $R_2 = 20R_1$, assuming the gain of the current mirror is unity.
- What is the GB of this amplifier?

$$GB = |A_v(0)|\omega_{-3dB} = \frac{R_2 A_o}{R_1(1 + A_o)} \frac{1}{R_2 C_o} = \frac{A_o}{(1 + A_o) R_1 C_o} = \frac{1}{2 R_1 C_o}$$

where C_o is the capacitance seen at the output of the current mirror.

- If $R_1 = 10 \text{ k}\Omega$ and $C_o = 250 \text{ fF}$, then $GB = 31.83 \text{ MHz}$.
- Limitations: $R_1 > R_{in} = 1/g_{m1}$ and $R_2 < r_{ds2} \parallel r_{ds6} \Rightarrow \frac{R_2}{R_1} \ll g_{m1}(r_{ds2} \parallel r_{ds6})$



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High GB Voltage Amplifier Design: Current Feedback

Design a wide-swing, cascode voltage amplifier to achieve a gain of -10 V/V and a GB of 500 MHz, which corresponds to a -3 dB frequency of 50 MHz.

Solution

- Because the desired gain is known, begin by assuming that C_o is 100 fF. To get a GB of 500 MHz, R_1 must be 3.2 k Ω and $R_2 = 32$ k Ω . So, $1/g_{m1}$ must be less than 3200 Ω (say 300 Ω). Therefore, you can write:

$$g_{m1} = \sqrt{2KI'(W/L)} = \frac{1}{300\Omega} \rightarrow 5.56 \times 10^{-6} = K' \cdot I \cdot \frac{W}{L} \rightarrow 0.0505 = I \frac{W}{L}$$

- At this point, there is a problem. If W/L is small to minimize C_o , the current is too high. If $W/L = 200 \mu\text{m}/1\mu\text{m}$, the current is 0.25 mA. However, using this W/L for M_4 and M_6 gives a value of C_o that is greater than 100 fF.
- Therefore, select $W/L = 200$ for M_1 , M_3 , M_5 and M_7 and $W/L = 20 \mu\text{m}/1 \mu\text{m}$ for M_2 , M_4 , M_6 , and M_8 , which gives a current in these transistors of 25 μA .
- Because R_2/R_1 is multiplied by 1/11, let R_2 be 110 times R_1 , or 352 k Ω .
- Now select a W/L for M_{12} of 20 $\mu\text{m}/1\mu\text{m}$, which enables you to calculate C_o .
- Assume zero bias on all voltage-dependent capacitors. Also, assume the diffusion area as 2 μm times the W . C_o can be written as:

$$C_o = C_{gd4} + C_{bd4} + C_{gd6} + C_{bd6} + C_{gs12}$$



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High GB Voltage Amplifier Design (continued)

The various capacitors are:

$$C_{gd4} = C_{gd6} = CGDO \times 10 \mu m = (220 \times 10^{-12}) (20 \times 10^{-6}) = 4.4 \text{ fF}$$

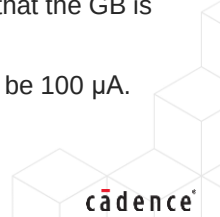
$$C_{bd4} = CJ \times AD_4 + CJSW \times PD_4 = (770 \times 10^{-6}) (20 \times 10^{-12}) + (380 \times 10^{-12}) (44 \times 10^{-6}) \\ = 15.4 \text{ fF} + 16.7 \text{ fF} = 32.1 \text{ fF}$$

$$C_{bd6} = (560 \times 10^{-6}) (20 \times 10^{-12}) + (350 \times 10^{-12}) (44 \times 10^{-6}) = 26.6 \text{ fF}$$

$$C_{gs12} = (220 \times 10^{-12} \cdot 20 \times 10^{-6}) + (0.67) (20 \times 10^{-6} \cdot 10^{-6} \cdot 24.7 \times 10^{-4}) = 37.3 \text{ fF}$$

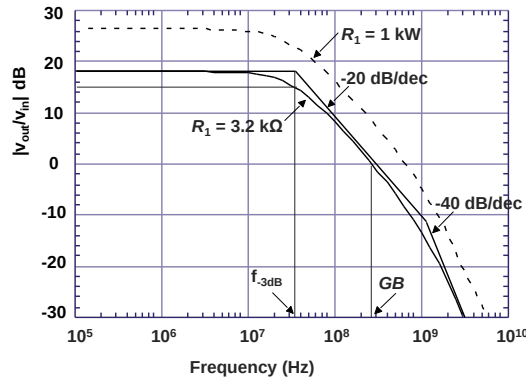
$$C_o = 4.4 \text{ fF} + 32.1 \text{ fF} + 4.4 \text{ fF} + 26.6 \text{ fF} + 37.3 \text{ fF} = 105 \text{ fF}$$

- Notice, if you did not reduce the W/L of M2, M4, M6, and M8, that Co would have
- easily exceeded 100 fF. Because 105 fF is close to the original guess of 100 fF, keep the values of R1 and R2. If this value was significantly different, then the values of R1 and R2 must be adjusted so that the GB is 500 MHz. You must also check to make sure that the input pole is greater than 500 MHz.
- Complete the design by assuming that IBias = 100 μ A and that the current in M9 through M12 be 100 μ A. Thus W13/L13 = W14/L14 = 20 μ m/1 μ m and W9/L9 through W12/L12 are 20 μ m/1 μ m.



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High GB Voltage Amplifier Design (continued)



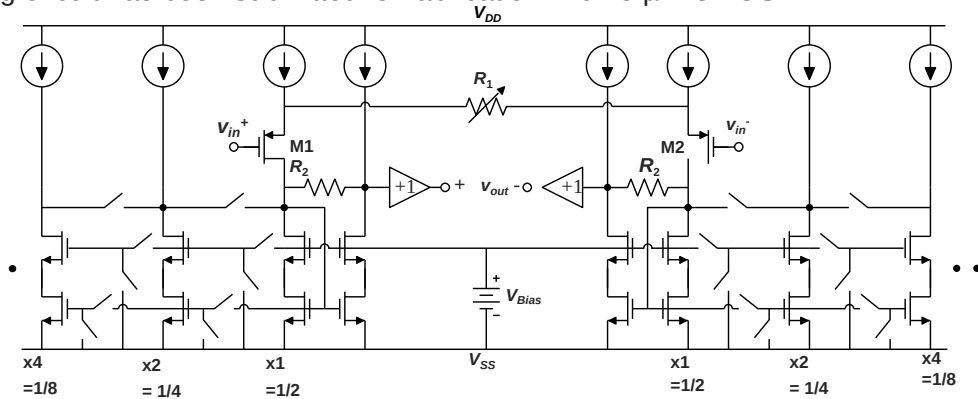
Simulation results:

- $f_{-3dB} \approx 38 \text{ MHz}$ $GB \approx 300 \text{ MHz}$
- Closed-loop gain = 18 dB (Loss of -2 dB is attributed to source follower and R_1 .)
- Notice second pole at about 1 GHz. To get these results, it was necessary to bias the input at -1.7 VDC using $\pm 3\text{V}$ power supplies.
- If R_1 is decreased to 1 k Ω , the results are Gain of 26.4 dB, $f_{-3dB} = 32 \text{ MHz}$, and $GB = 630 \text{ MHz}$

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71 MHz PGA: Current Amplifier

The following circuit has been submitted for fabrication in 0.25 μm CMOS.

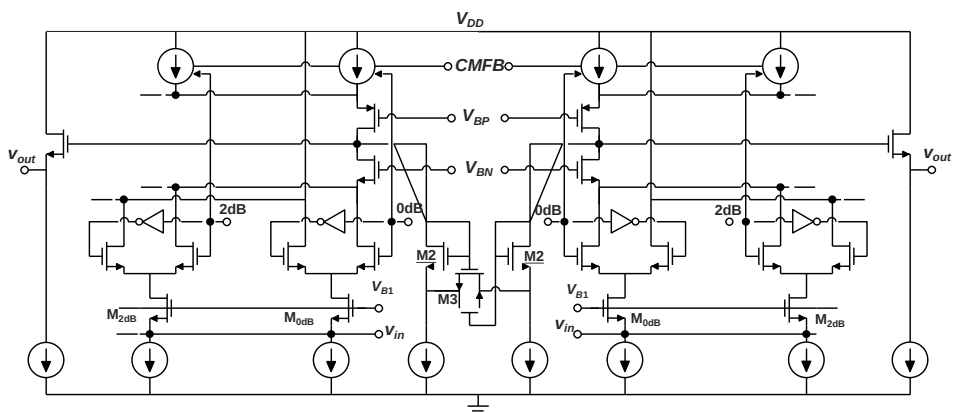


R_1 and the current mirrors are used for gain variation. R_2 is fixed. This amplifier can be cascaded for higher gains.

$$BW = BW_i \sqrt{2^{1/n} - 1} \quad \text{for } n = 2, \quad BW = 0.64 BW_i$$

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71 MHz CMOS Programmable Gain Amps



- $R_{in} = 330\Omega$ to match source driving requirement.
- All current sinks are identical for the differential switches.
- Dominant pole at 150 MHz.

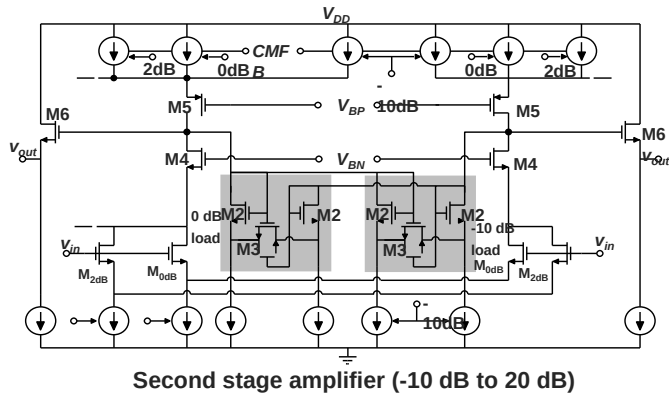
P. Orsatti, F. Piazza, and Q. Huang, "A 71 MHz CMOS IF-Basdband Strip for GSM," *IEEE Journal Solid-State Circuits*, vol. 35, No. 1, Jan. 2000, pp. 104-108.

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71 MHz CMOS Programmable Gain Amps (continued)

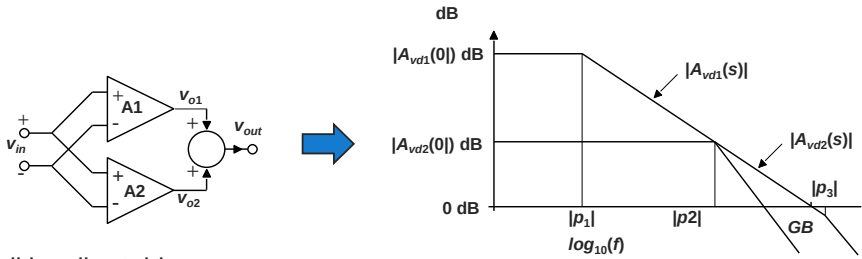


- Dominant pole is also at 150 MHz.
- For $V_{DD} = 2.5V$, at 60 dB gain, the total current is 2.6 mA.
- $IIP_3 \approx +1dBm$

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Parallel-Path Opamps

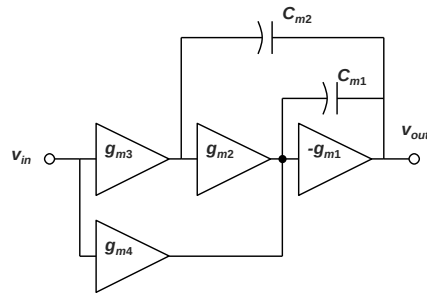
Parallel-path opamps combine high-gain, low-frequency paths with low-gain, high-frequency paths.



- Opamp is conditionally stable.
- Compensation is challenging.

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Multipath Nested Miller Compensation



Example of multipath nested Miller compensation

- All Miller capacitances must be around inverting stages.
- Ensure that the RHP zeros that are generated by the Miller compensation are canceled.
- Avoid pole-zero doublets, which can introduce a slow time constant.

R.G.H. Eschauzier and J.H.Huijsing, *Frequency Compensation Techniques for Low-Power Operational Amplifiers*, Kluwer Academic publishers, 1995, Chapter 6.

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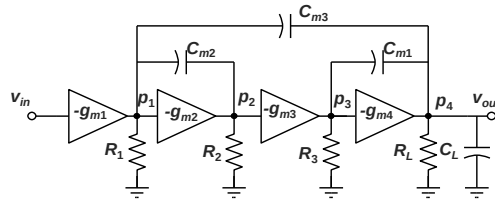
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Hybrid Nested Miller Compensation

Compensating results

- C_{m1} pushes p_4 to higher frequencies and p_3 down to lower frequencies.
- C_{m2} pushes p_2 to higher frequencies and p_1 down to lower frequencies.
- C_{m3} pushes p_3 to higher frequencies (feedback path) and pulls p_1 further to lower frequencies.



Equations

$$GB \approx g_{m1} / C_{m3} \quad p_2 \approx g_{m2} / C_{m3} \quad p_3 \approx g_{m3} C_{m3} / C_{m1} C_{m2} \quad p_4 \approx g_{m4} / C_L$$

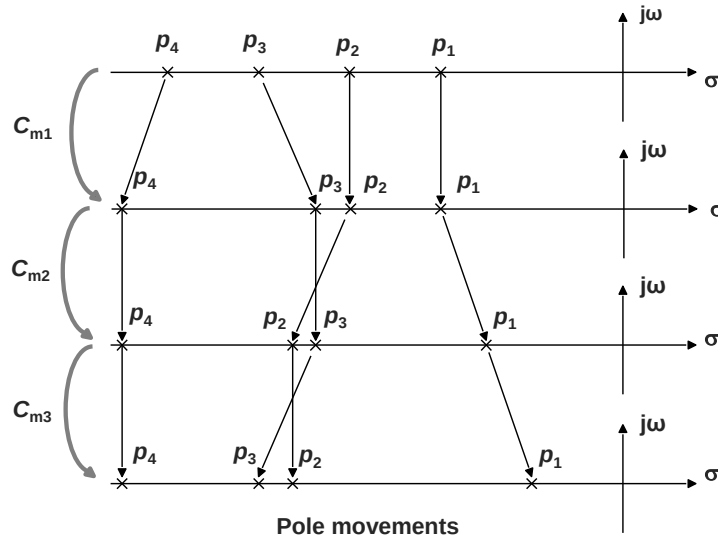
Design

$$GB < p_2, p_3, p_4$$

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Hybrid Nested Miller Compensation Technique

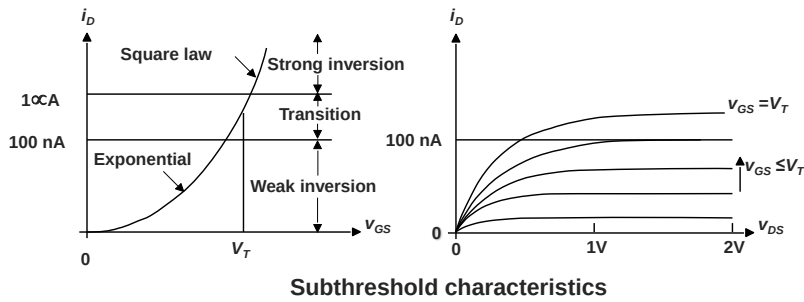
The figure shows pole movements as a function of compensation capacitances.



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Subthreshold Operation

Most micropower opamps use transistors in the subthreshold region.

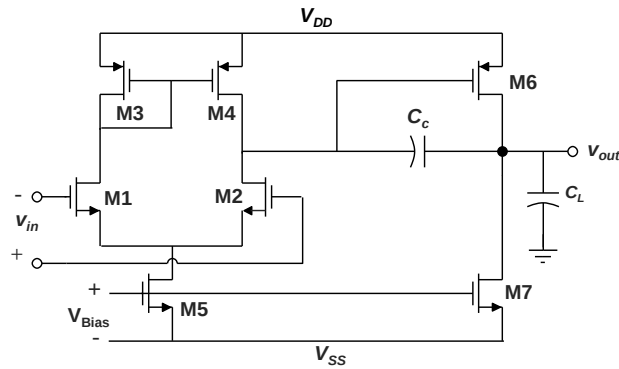


$$i_D = \frac{W}{L} I_{DO} \exp\left(\frac{qV_{GS}}{nkT}\right)(1 + \lambda v_{DS}) \Rightarrow g_m = \frac{qI_D}{nkT} \text{ and } g_{ds} \approx \lambda I_D$$

Operation with channel length = $L_{\min}$ also is normally in weak inversion.

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Two-Stage Miller Opamp: Weak Inversion



Low-frequency response:

$$A_{vo} = g_{m2}g_{m6} \left(\frac{r_{o2}r_{o4}}{r_{o2} + r_{o4}} \right) \left(\frac{r_{o6}r_{o7}}{r_{o6} + r_{o7}} \right) = \frac{1}{n_2 n_6 (kT/q)^2 (\lambda_2 + \lambda_4)(\lambda_6 + \lambda_7)} \quad \left(\text{No longer } \propto \frac{1}{\sqrt{I_D}} \right)$$

$$\text{GB and SR: } GB = \frac{I_{D1}}{(n_1 kT/q)C} \text{ and } SR = \frac{I_{D5}}{C} = 2 \frac{I_{D1}}{C} = 2GB \left(n_1 \frac{kT}{q} \right) = 2GBn_1 V_t$$

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Example: Gain and GB for Subthreshold Opamp

1. Calculate the gain, GB , and SR of the opamp shown on the previous slide.

The currents are $I_{D5} = 200 \text{ nA}$ and $I_{D7} = 500 \text{ nA}$. The device lengths are $1 \text{ }\mu\text{m}$. Values for n are 1.5 and 2.5 for p-channel and n-channel transistors, respectively. The compensation capacitor is 5 pF . Assume that the temperature is 27°C .

2. If $V_{DD} = 1.5\text{V}$ and $V_{SS} = -1.5\text{V}$, what is the power dissipation of this opamp?

Solution

The low-frequency small-signal gain is

$$A_v = \frac{1}{(1.5)(2.5)(0.026)^2(0.04 + 0.05)(0.04 + 0.05)} = 43,701 \text{ V/V}$$

The gain bandwidth is $GB = \frac{100 \times 10^{-9}}{2.5(0.026)(5 \times 10^{-12})} = 307,690 \text{ rad/sec} \cong 49.0 \text{ kHz}$

The slew rate is $SR = (2)(307,690)(2.5)(0.026) = 0.04 \text{ V}/\mu\text{s}$

The power dissipation is $P_{diss} = 3(0.7 \mu\text{A}) = 2.1 \mu\text{W}$



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Push-Pull Output Opamp in Weak Inversion

First stage gain is

$$A_{vo} = \frac{g_{m2}}{g_{m4}} = \frac{I_{D2}n_4V_t}{I_{D4}n_2V_t} = \frac{I_{D2}n_4}{I_{D4}n_2} \cong 1$$

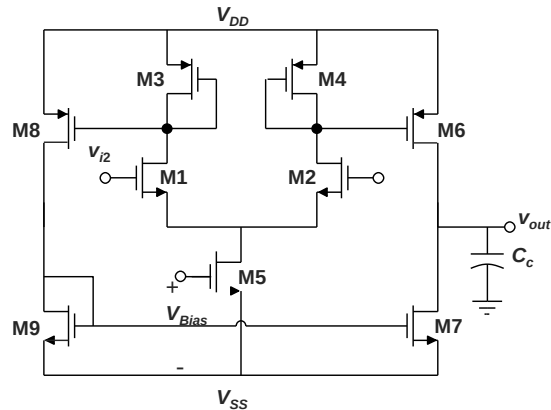
Total gain is

$$A_{vo} = \frac{g_{m1}(S_6/S_4)}{(g_{ds6} + g_{ds7})} = \frac{(S_6/S_4)}{(\lambda_6 + \lambda_7)n_1V_t}$$

At room temperature ($V_t = 0.0259V$) and for typical device lengths, gains of 60 dB can be obtained.

The GB is

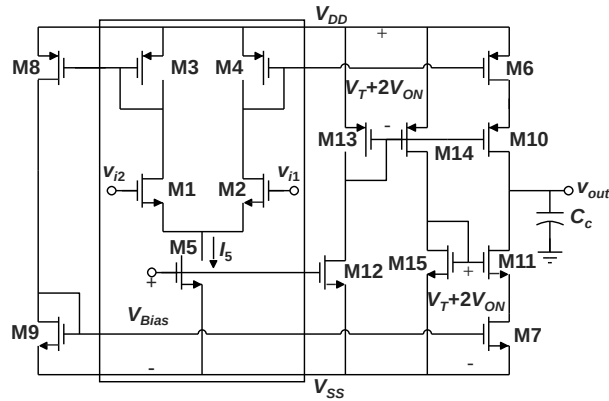
$$GB = \frac{g_{m1}}{C} \left(\frac{S_6}{S_4} \right) = \frac{g_{m1}b}{C}$$



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Increasing Gain of Previous Opamp

1. Reduce the currents in M3 and M4 and introduce gain in the current mirrors.
2. Use a cascode output stage. (Cannot use self-biased cascode because currents are too low).



You can easily achieve gains greater than 80 dB with a power dissipation of less than 1μW.

$$A_v = \left(\frac{g_{m1} + g_{m2}}{2} \right) R_{out} = \frac{g_{m1}}{\frac{g_{ds6}g_{ds10}}{g_{m10}} + \frac{g_{ds7}g_{ds11}}{g_{m11}}} = \frac{\frac{I_5}{2n_nV_t}}{\frac{I_7^2\lambda_n^2}{n_nV_t} + \frac{I_7^2\lambda_p^2}{n_nV_t}} = \left(\frac{I_5}{2I_7} \right) \left(\frac{1}{n_nV_t^2(n_n\lambda_n^2 + n_p\lambda_p^2)} \right)$$

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Increasing Output Current (continued)

- How much output current is available from this circuit if there is no current gain from the input to output stage?
- Assume transistors M18 through M21 are equal to M3 and M4 and that transistors M22 through M27 are all equal.

$$\frac{W_{28}}{L_{28}} = A \left(\frac{W_{26}}{L_{26}} \right) \text{ and } \frac{W_{29}}{L_{29}} = A \left(\frac{W_{27}}{L_{27}} \right)$$

- The output current available can be found by assuming that $v_{in} = v_{i1} - v_{i2} > 0$

$$\therefore i_1 + i_2 = I_5 + A(i_2 - i_1)$$

- The ratio of i_2 to i_1 can be expressed as $\frac{i_2}{i_1} = \exp\left(\frac{v_{in}}{nV_t}\right)$

- Defining the output current as $i_{OUT} = b(i_2 - i_1)$ and combining the two equations gives

$$i_{OUT} = \frac{bI_5 \left[\exp\left(\frac{v_{in}}{nV_t}\right) - 1 \right]}{(1 + A) - (A - 1)\exp\left(\frac{v_{in}}{nV_t}\right)} \Rightarrow i_{OUT} = \infty \text{ when } A = 2.16 \text{ and } \frac{v_{in}}{nV_t} = 1$$

where b corresponds to any current gain through the current mirrors (M5-M4 and M8-M3).



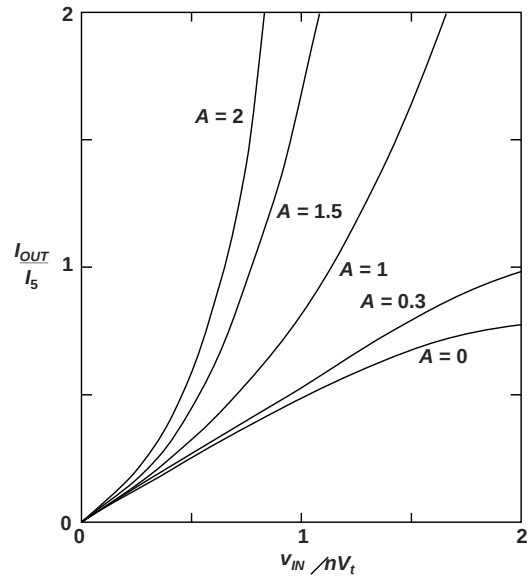
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Dynamically Biased Differential Amplifier: Overdrive

- The enhanced output current is accomplished by the use of positive feedback (M28-M2-M19-M28).
- The loop gain is

$$LG = \left(\frac{g_{m28}}{g_{m4}} \right) \left(\frac{g_{m19}}{g_{m26}} \right) = A \frac{g_{m19}}{g_{m4}} = A$$

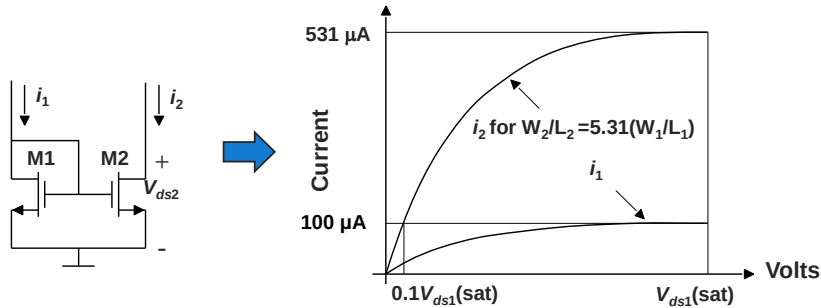
- Notice that as the output current increases, the transistors leave the weak inversion region, and the previous analysis is no longer valid.



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Increasing the Output Current: Strong Inversion Operation

A useful technique is to bias the output transistor of a current mirror in the active region and then, during large overdrive, cause the output transistor to become saturated, causing a significant current gain.



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Example: Current Mirror with M2 in Active Region

Assume that M2 has a voltage across the drain-source of $0.1V_{ds(sat)}$:

- Design the W_2/L_2 ratio so that $I_1 = I_2 = 100 \mu A$ if $W_1/L_1 = 10$.
- Find the value of I_2 if M2 is saturated.

Solution

Assume that the saturation voltage of M2 is

$$V_{ds1(sat)} = \sqrt{\frac{2I_1}{K_N'(W_1/L_1)}} = \sqrt{\frac{200}{110 \cdot 10}} = 0.4264V$$

Now using the active equation of M2, we set $I_2 = 100 \mu A$ and solve for W_2/L_2 .

$$\begin{aligned} 100 \mu A &= K_N'(W_2/L_2) \left[V_{ds1(sat)} \cdot V_{ds2} - 0.5 V_{ds2}^2 \right] \\ &= 110 \mu A/V^2 (W_2/L_2) \left[0.426 \cdot 0.0426 - 0.5 \cdot 0.0426^2 \right] V^2 = 1.883 \times 10^6 (W_2/L_2) \end{aligned}$$

Thus,

$$100 = 1.883 (W_2/L_2) \rightarrow \frac{W_2}{L_2} = 53.12$$

Now if M2 becomes saturated, the value of the output current of the mirror with $100 \mu A$ input will be $531 \mu A$ or a boosting of 5.31 times I_1 .



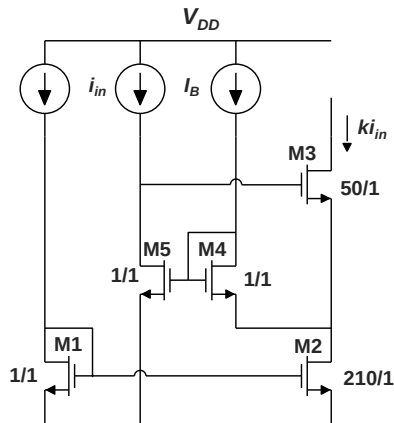
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Achieving Current Mirror Boosting: A Better Way

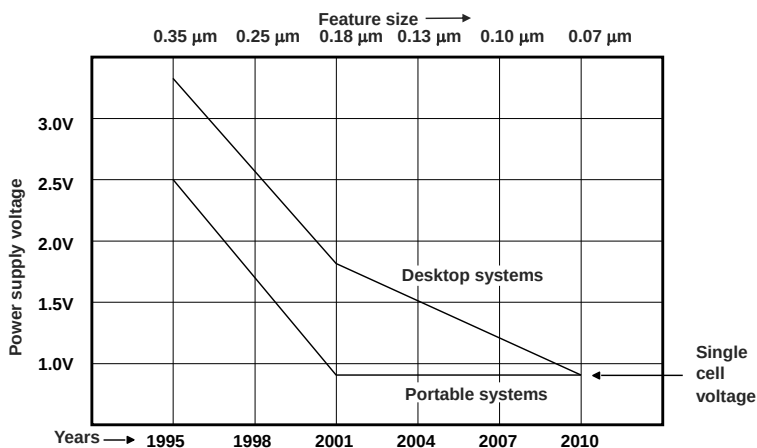
The current mirror boosting concept that was illustrated on the previous slide suffers from the limitation that an increase in the current through the cascode device (M16) increases V_{GS16} and thereby limits the increase of V_{DS12} . This can be overcome by the following circuit.



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SIA Road Map for Power Supplies

- The following Semiconductor Industry Association (SIA) graph shows that over time the power supply voltage is dropping as the feature size becomes smaller.



- While low-voltage opamps can be easily designed in weak inversion, strong.
- inversion leads to higher performance and is the focus of this section. Threshold voltages remain about 0.5 to 0.7V to enable the MOSFET to be turned off.

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Low-Voltage Strong-Inversion Operation

- Reduced power supply means decreased dynamic range.
- Nonlinearity increases because the transistor is working close to $V_{DS}(\text{sat})$.
- Large values of λ exist because the transistor is working close to $V_{DS}(\text{sat})$.
- Drain-bulk and source-bulk capacitances are increased because they are less reverse biased.
- Large values of currents and W/L ratios enable high transconductance.
- Small values of currents and large values of W/L give small $V_{DS}(\text{sat})$.
- There is a severely reduced input common-mode range.
- Switches require charge pumps.



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Low-Voltage Strong-Inversion Operation (continued)

Approach

- Low-voltage input stages with reasonable ICMR
- Low-voltage bias and load circuits
- Low-voltage opamps



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Differential Amplifier with Current Source Loads

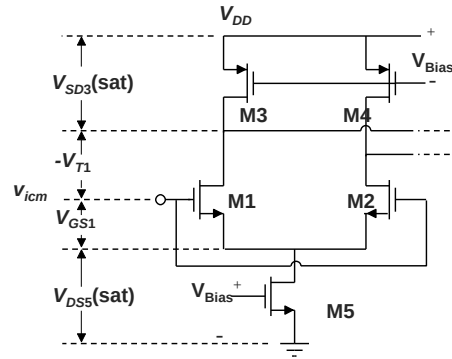
Minimum power supply (ICMR = 0)

$$\begin{aligned} V_{DD}(\min) &= V_{SD3}(\text{sat}) - V_{T1} + V_{GS1} + V_{DS5}(\text{sat}) \\ &= V_{SD3}(\text{sat}) + V_{DS1}(\text{sat}) + V_{DS5}(\text{sat}) \end{aligned}$$

Input common-mode range

$$V_{icm}(\text{upper}) = V_{DD} - V_{SD3}(\text{sat}) + V_{T1}$$

$$V_{icm}(\text{lower}) = V_{DS5}(\text{sat}) + V_{GS1}$$



Example: If the threshold magnitudes are 0.7V, $V_{DD} = 1.5\text{V}$ and the saturation voltages are 0.3V, then

$$V_{icm}(\text{upper}) = 1.5 - 0.3 + 0.7 = 1.9\text{V} \quad \text{and} \quad V_{icm}(\text{lower}) = 0.3 + 1.0 = 1.3\text{V}$$

giving an ICMR of 0.6V.

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Increasing ICMR with Parallel Input Stages

Turn-on voltage for the n-channel input:

$$V_{onn} = V_{DSN5}(sat) + V_{GSN1}$$

Turn-on voltage for the p-channel input:

$$V_{onp} = V_{DD} - V_{SDP5}(sat) - V_{SGP1}$$

The sum of V_{onn} and V_{onp} equals the minimum power supply.

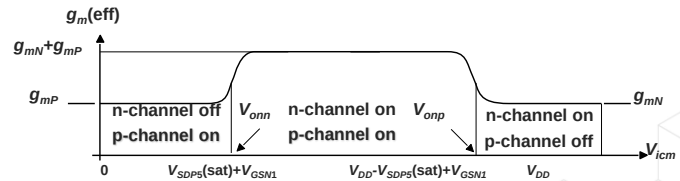
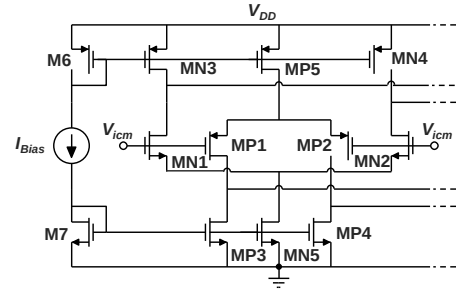
Regions of operation:

$$V_{DD} > V_{icm} > V_{onp} : \quad (\text{n-channel on and p-channel off}) \quad g_m(eq) = g_{mN}$$

$$V_{onp} \geq V_{icm} \geq V_{onn} : \quad (\text{n-channel on and p-channel on}) \quad g_m(eq) = g_{mN} + g_{mP}$$

$$V_{onn} > V_{icm} > 0 : \quad (\text{n-channel off and p-channel on}) \quad g_m(eq) = g_{mP}$$

where $g_m(eq)$ is the equivalent input transconductance of the input stage, g_{mN} is the input transconductance for the n-channel input and g_{mP} is the input transconductance for the p-channel input.



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Removing the Nonlinearity in Transconductances

Increase the bias current in the differential amplifier that is on when the other differential amplifier is off.

Three regions of operation depending on the value of V_{icm} :

1. $V_{icm} < V_{onn}$: n-channel diff amp off and p-channel on with $I_p = 4I_b$:

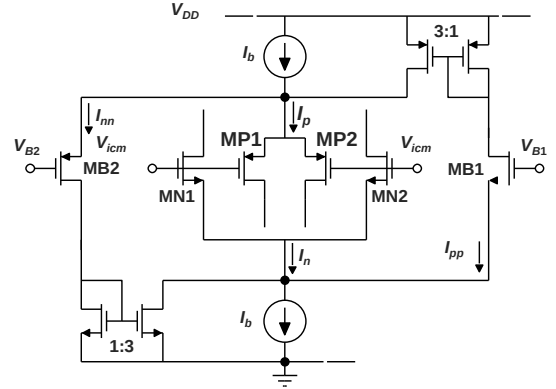
$$g_m(eff) = \sqrt{\frac{K_p' W_p}{L_p}} 2\sqrt{I_b}$$

2. $V_{onn} < V_{icm} < V_{onp}$: both on with $I_n = I_p = I_b$:

$$g_m(eff) = \sqrt{\frac{K_N' W_N}{L_N}} \sqrt{I_b} + \sqrt{\frac{K_P' W_P}{L_P}} \sqrt{I_b}$$

3. $V_{icm} > V_{onp}$: p-channel diff amp off and n-channel on with $I_n = 4I_b$:

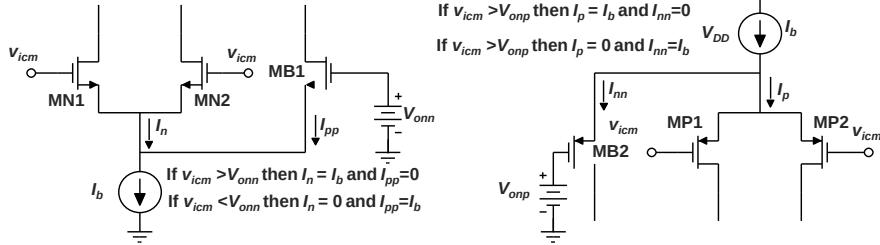
$$g_m(eff) = \sqrt{\frac{K_N' W_N}{L_N}} 2\sqrt{I_b}$$



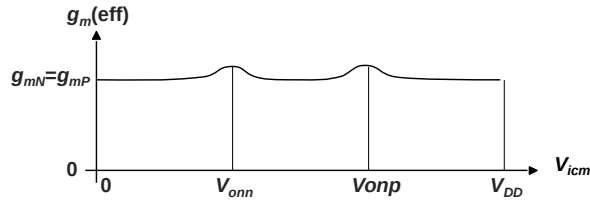
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How Does the Current Compensation Work?

Set $V_{B1} = V_{onn}$ and $V_{B2} = V_{onp}$



Result:

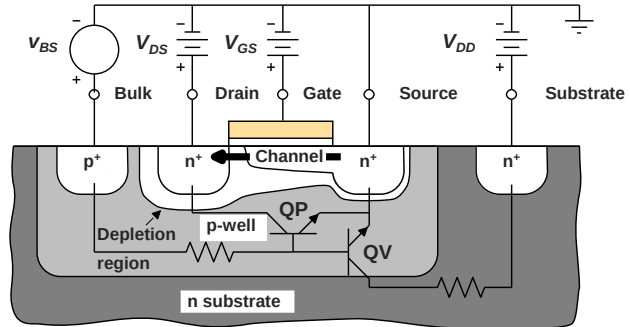


These techniques and many similar ones are good for power-supply values down to about 1.5V. Less than that, different techniques must be used, or the technology must be modified (natural devices).

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Bulk-Driven MOSFET

- A depletion device would permit large ICMR even with very small power supply voltages because V_{GS} is zero or negative.
- When a MOSFET is driven from the bulk with the gate held constant, it acts like a depletion transistor.



Cross section of an n-channel bulk-driven MOSFET

- Large-signal equation:

$$i_D = \frac{k_n' W}{2L} \left[V_{GS} - V_{T0} - \gamma \sqrt{2|\phi_F| - V_{BS}} + \gamma \sqrt{2|\phi_F|} \right]^2$$

- Small-signal transconductance:

$$g_{mbs} = \frac{\gamma \sqrt{(2k_n' W / L) I_D}}{2\sqrt{2|\phi_F| - V_{BS}}}$$

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Bulk-Driven MOSFET (continued)

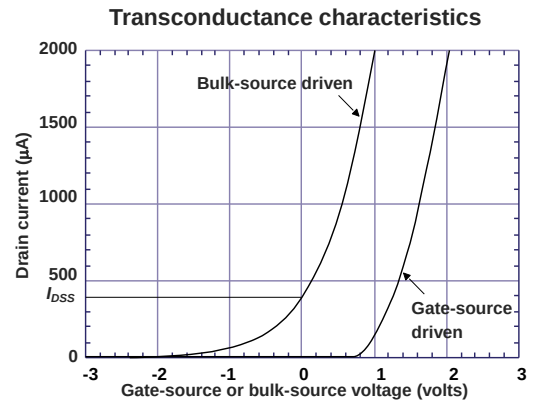
Saturation

$$V_{DS} > V_{BS} - V_P$$

$$V_{BS} = V_P + V_{ON}$$

$$i_D = I_{DSS} \left(1 - \frac{V_{BS}}{V_P} \right)^2$$

- $g_m(\text{bulk}) > g_m(\text{gate})$ if $V_{BS} > 0$ (forward biased).
- Noise of both configurations are the same (any differences comes from the gate versus bulk noise).
- Bulk-driven MOSFET tends to be more linear at lower currents than the gate-driven MOSFET.
- Very useful for generation of I_{DSS} floating current sources.



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Bulk-Driven N-channel Differential Amplifier

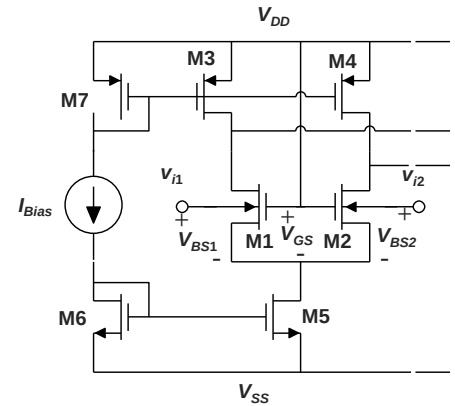
What is the ICMR? $V_{icm}(\min) = V_{SS} + V_{DS5}(\text{sat}) + V_{BS1} = V_{SS} + V_{DS5}(\text{sat}) - |V_{P1}| + V_{DS1}(\text{sat})$
 $|V_{P1}| > V_{DS5}(\text{sat}) + V_{DS1}(\text{sat})$

Notice that V_{icm} can be less than V_{SS} if:

$$V_{icm}(\max) = ?$$

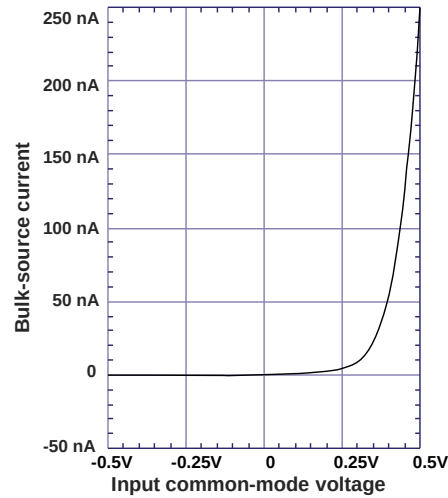
- As V_{icm} increases, the current through M1 and M2 is constant so the source increases.
- However, the gate voltage stays constant so that V_{GS1} decreases.
- Because the current must remain constant through M1, and M2 because of M5, the bulk-source voltage becomes less negative.
- This causes V_{TN1} to decrease and maintain the currents through M1 and M2 constant.
- If V_{icm} is increased sufficiently, the bulk-source voltage becomes positive.
- However, current does not start to flow until V_{BS} is greater than 0.3 volts, so the effective $V_{icm}(\max)$ is:

$$V_{icm}(\max) \approx V_{DD} - V_{SD3}(\text{sat}) - V_{SD1}(\text{sat}) + V_{BS1}$$



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ICMR of the Bulk-Driven, Differential Amplifier

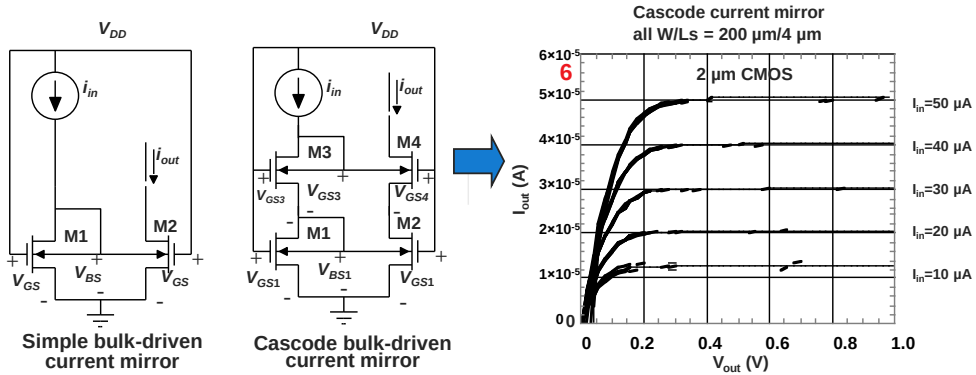


- Effective ICMR is from V_{SS} to $V_{DD} - 0.3V$.
- The transconductance of the input stage can vary as much as 100% over the ICMR, which makes it very difficult to compensate.

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Low-Voltage Current Mirrors with Bulk-Driven MOSFET

- The biggest problem with current mirrors is the large minimum input voltage required for previously examined current mirrors.
- If the bulk-driven MOSFET is biased with a current that exceeds I_{DSS} , then it is enhancement and can be used as a current mirror.

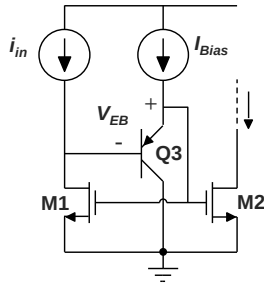


- The cascode current mirror gives a minimum input voltage of less than 0.5V for currents less than 100 μA .

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Simple Current Mirror with Level Shifting

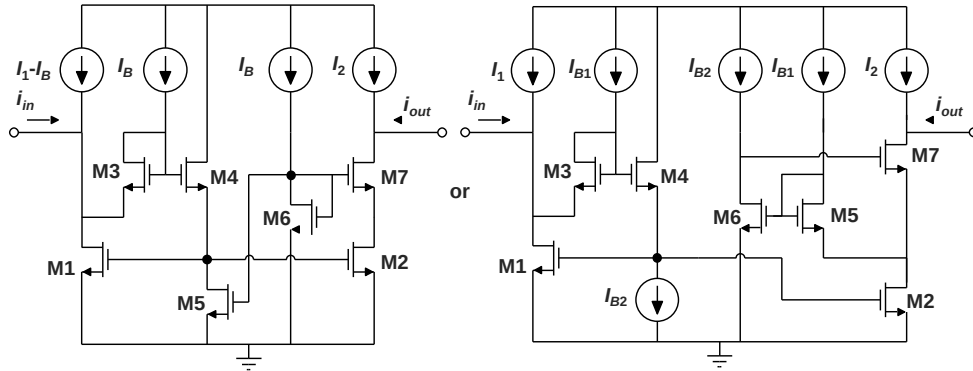
Because the drain can be V_T less than the gate, the drain can be biased to reduce the minimum input voltage as shown.



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Low-Voltage Current Mirror: Wide Input/Output Swings

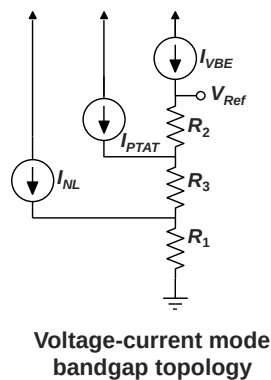
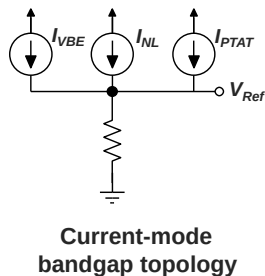
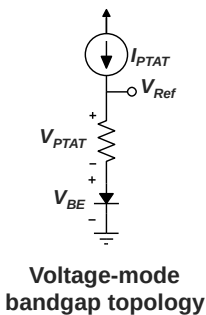
The current mirror shown requires a power supply of $V_T + 3V_{ON}$ and has a $V_{in(min)} = V_{ON}$ and a $V_{out(min)} = 2V_{ON}$ (less for the regulated cascode output mirror).



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Bandgap Topologies and Low-Voltage Power Supply

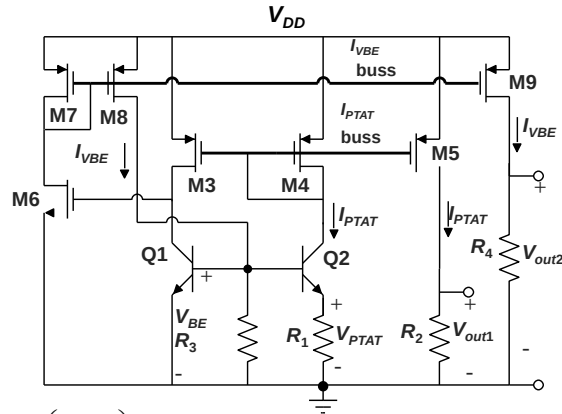
Following are bandgap topologies that are compatible with low-voltage power supplies:



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Currents with VBE and PTAT Temperature Coefficients

Following is a method of generating currents with VBE and PTAT temperature coefficients:



$$V_{out1} = I_{PTAT} R_2 = \left(\frac{V_{PTAT}}{R_1} \right) R_2 = V_{PTAT} \frac{R_2}{R_1}$$

$$V_{out2} = I_{VBE} R_4 = \left(\frac{V_{BE}}{R_3} \right) R_4 = V_{BE} \frac{R_4}{R_3}$$

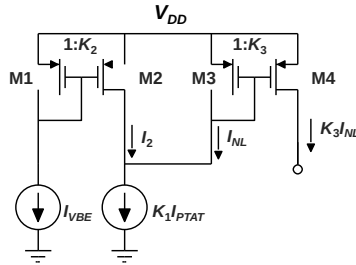
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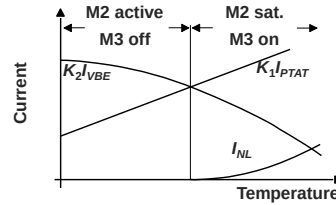


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Technique: Canceling the Bandgap Curvature



Circuit to generate nonlinear correction term, I_{NL}



Various currents

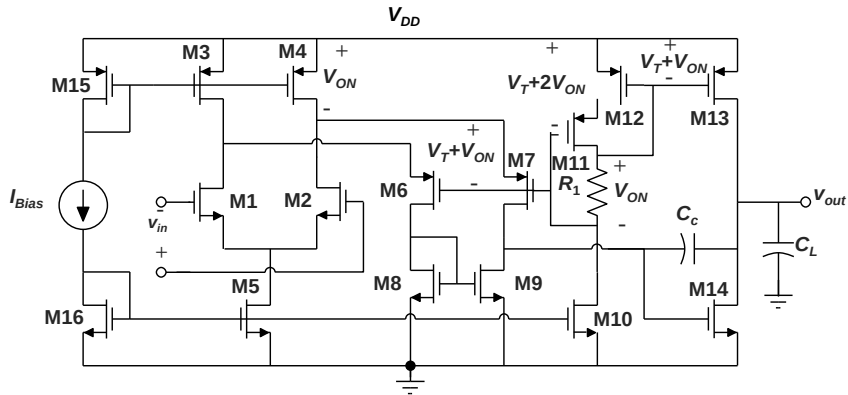
$$I_{NL} = \begin{cases} 0, & K_2 I_{VBE} > K_1 V_{PTAT} \\ K_1 V_{PTAT} - K_2 I_{VBE}, & K_2 I_{VBE} < K_1 V_{PTAT} \end{cases}$$

The combination of this concept with the previous slide yielded a curvature corrected bandgap reference of 0.596V with a TC of 20 ppm/C° from -15 C° to 90 C° using a 1.1V power supply. In addition, the line regulation was 408 ppm/V for $1.2 \leq V_{DD} \leq 10V$ and 2000 ppm/V for $1.1 \leq V_{DD} \leq 10V$. The quiescent current was 14 μA .

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Classical Techniques: Low-Voltage Opamps

Following is a low-voltage opamp using a classical technique ($V_{DD} \geq 2V_T$):



- Clever use of classical techniques
- Balanced inputs

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Classical Design: Low-Voltage Opamps

Design the opamp from the previous slide to meet these specifications:

$$V_{DD} = 2V$$

$$V_{icm}(\max) = 2.5V$$

$$V_{icm}(\min) = 1V$$

$$V_{out}(\max) = 1.75V$$

$$V_{out}(\min) = 0.5V$$

$$GB = 10 \text{ MHz}$$

$$\text{Slew rate} = \pm 10 \text{ V}/\mu\text{s}$$

$$\text{Phase margin} = 60^\circ \text{ for } C_L = 10 \text{ pF}$$

Solution

Assuming the conditions for a two-stage opamp necessary to achieve 60° C phase margin and that the RHP zero is at least 10 GB gives $C_c = 0.2C_L = 2 \text{ pF}$

$$I_5 = C_c \cdot SR = 2 \times 10^{-12} \cdot 10^7 = 20 \mu\text{A}$$

The slew rate is directly related to the current in M5 and gives:

$$g_{m1} = g_{m2} = GB \cdot C_c = 20\pi \times 10^6 \cdot 2 \times 10^{-12} = 125.67 \mu\text{S}$$

From GB and C_c , the input transconductances are:

$$\frac{W_1}{L_1} = \frac{W_2}{L_2} = \frac{g_{m1}^2}{2k_n'(I_1/2)} = \frac{(125.67 \times 10^{-6})^2}{2 \cdot 110 \times 10^{-6} \cdot 10 \times 10^{-6}} = 7.18$$

Knowing the current flow in M1 and M2 gives the W/L ratios as



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Design: Low-Voltage Opamps (continued)

- Next, you find the W/L of M5 that will satisfy $V_{icm}(\min)$ specification.

$$V_{icm}(\min) = V_{DS5}(sat) + V_{GS1}(10\mu A) = 1V$$

This gives:

$$V_{DS5}(sat) = 1 - \sqrt{\frac{2 \cdot 10}{110 \cdot 7.18}} - 0.75 = 1 - 0.159 - 0.75 = 0.0909V$$

$$\therefore V_{DS5}(sat) = 0.0909 = \sqrt{\frac{2 \cdot I_5}{k_n'(W_5/L_5)}} \rightarrow \frac{W_5}{L_5} = \frac{2 \cdot 20}{110 \cdot (0.0909)^2} = 44$$

- The design of M3 and M4 is accomplished from the upper-input common-mode voltage:

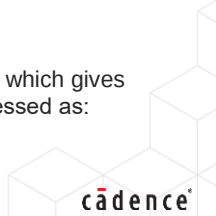
$$V_{icm}(\max) = V_{DD} - V_{DS3}(sat) + V_{TN} = 2 - V_{DS3}(sat) + 0.75 = 2.5V$$

- Solving for $V_{SD3}(sat)$ gives 0.25V. Assume the currents in M6 and M7 are 20 μA . This gives a current of 30 μA in M3 and M4. Knowing the current in M3 (M4) gives:

$$V_{DS3}(sat) \leq \sqrt{\frac{2 \cdot 30}{50 \cdot (W_3/L_3)}} \rightarrow \frac{W_3}{L_3} = \frac{W_4}{L_4} \geq \frac{2 \cdot 30}{(0.25)^2 \cdot 50} = 19.2$$

- Next, using the $V_{SD}(sat) = V_{ON}$ of M3 and M4, design M10 through M12. Also assume that $I_{10} = I_5 = 20 \mu A$, which gives $W_{10}/L_{10} = 44$. R_1 is designed as $R_1 = 0.25 V / 20 \mu A = 12.5 k\Omega$. The W/L ratios of M11 and M12 can be expressed as:

$$\frac{W_{11}}{L_{11}} = \frac{W_{12}}{L_{12}} = \frac{2 \cdot I_{11}}{k_p' V_{SD11}(sat)^2} = \frac{2 \cdot 20}{50 \cdot (0.25)^2} = 12.8$$



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Design: Low-Voltage Opamps (continued)

- Because the source-gate voltages and currents of M6 and M7 are the same as M11 and M12 then the W/L values are equal. Thus:

$$W_6 / L_6 = W_7 / L_7 = 12.8$$

- M8 and M9 should be as small as possible to reduce the parasitic (mirror) pole. However, the voltage drop across M4, M6 and M8 must be less than the power supply. Using this to design the gate-source voltage of M8 gives. Thus:

$$V_{GS8} = V_{DD} - 2V_{ON} = 2V - 2 \cdot 0.25 = 1.5V$$

$$\frac{W_8}{L_8} = \frac{W_9}{L_9} = \frac{2 \cdot I_8}{k_n' V_{DS8} (sat)^2} = \frac{2 \cdot 30}{110 \cdot (0.75)^2} = 0.97 \approx 1$$

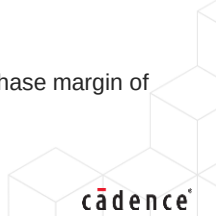
Because M8 and M9 are small, the mirror pole is insignificant.

- The next poles of interest would be those at the sources of M6 and M7. Assuming the channel length is 1 μm , these poles are given as

$$p_6 \approx \frac{g_{m6}}{C_{GS6}} = \frac{\sqrt{2k_p' (W_6 / L_6) \cdot I_6}}{(2/3) \cdot W_6 \cdot L_6 \cdot C_{ox}} = \frac{\sqrt{2 \cdot 50 \cdot 12.8 \cdot 20 \times 10^{-6}}}{(2/3) \cdot 12.8 \cdot 1 \cdot 2.47 \times 10^{-15}} = 7.59 \times 10^9 \text{ rads/sec}$$

which is about 100 times greater than GB.

- Finally, the W/L ratios of the second stage must be designed. You can either use the relationship for 60° phase margin of $g_{m14} = 10 g_{m1} = 1256.7 \mu\text{S}$ or consider proper mirroring between M9 and M14.



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Design: Low-Voltage Opamps (continued)

Substituting $1256.7 \mu\text{S}$ for g_{m14} and 0.5V for V_{DS14} in $W/L = g_m / (k_n' V_{DS}(\text{sat}))$ gives $W_{14}/L_{14} = 22.85$ which gives $I_{14} = 314 \mu\text{A}$. The W/L of M13 is designed by the necessary current ratio desired between the two transistors and is

$$\frac{W_{13}}{L_{13}} = \frac{I_{13}}{I_{12}} I_{12} = \frac{314}{20} \cdot 12.8 = 201$$

Now, check to make sure that the $V_{out}(\text{max})$ is satisfied. The saturation voltage of M13 is:

$$V_{SD13}(\text{sat}) = \sqrt{\frac{2 \cdot I_{13}}{k_p' (W_{13}/L_{13})}} = \sqrt{\frac{2 \cdot 314}{50 \cdot 201}} = 0.25\text{V}$$

which exactly meets the specification. For proper mirroring, the W/L ratio of M14 is:

$$\frac{W_9}{L_9} = \frac{I_9}{I_{14}} \frac{W_{14}}{L_{14}} = 1.46$$

Since W_9/L_9 was selected as 1, this is close enough.

The parameters are $g_{ds7} = 1 \mu\text{S}$, $g_{ds8} = 0.8 \mu\text{S}$, $g_{ds13} = 15.7 \mu\text{S}$ and $g_{ds14} = 12.56 \mu\text{S}$.

Therefore, small signal voltage gain is $(R_i \approx r_{ds9})$ because M7 is part of a cascode conf.).

$$\frac{v_{out}}{v_{in}} \approx \left(\frac{g_{m1}}{g_{ds9}} \right) \left(\frac{g_{m14}}{g_{ds13} + g_{ds14}} \right) = \left(\frac{125.6}{1.8} \right) \left(\frac{1256.7}{28.26} \right) = 69.78 \cdot 44.47 = 3,103 \text{V/V}$$

The power dissipation, including I_{bias} of $20 \mu\text{A}$, is $708 \mu\text{W}$.

The minimum power supply voltage is $V_T + 3\Delta V \approx 1.5\text{V}$ if $V_T = 0.7\text{V}$ and $\Delta V \approx 0.25\text{V}$.



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Performance of 1V Two-Stage Opamps

Specification ($V_{DD}=0.5V$, $V_{SS}=-0.5V$)	Measured Performance ($C_L = 22pF$)
DC open-loop gain	49dB (V_{icm} mid range)
Power supply current	300 μ A
Unity-gainbandwidth (GB)	1.3MHz (V_{icm} mid range)
Phase margin	57° (V_{icm} mid range)
Input offset voltage	$\pm 3mV$
Input common mode voltage range	-0.475V to 0.450V
Output swing	-0.475V to 0.491V
Positive slew rate	+0.7V/ μ sec
Negative slew rate	-1.6V/ μ sec
THD, closed loop gain of -1V/V	-60dB (0.75Vp-p, 1kHz sinewave) -59dB (0.75Vp-p, 10kHz sinewave)
THD, closed loop gain of +1V/V	-59dB (0.75Vp-p, 1kHz sinewave) -57dB (0.75Vp-p, 10kHz sinewave)
Spectral noise voltage density	367nV/ $\sqrt{Hz}$ @ 1kHz 181nV/ $\sqrt{Hz}$ @ 10kHz, 81nV/ $\sqrt{Hz}$ @ 100kHz 444nV/ $\sqrt{Hz}$ @ 1MHz
Positive Power Supply Rejection	61dB at 10kHz, 55dB at 100kHz, 22dB at 1MHz
Negative Power Supply Rejection	45dB at 10kHz, 27dB at 100kHz, 5dB at 1MHz

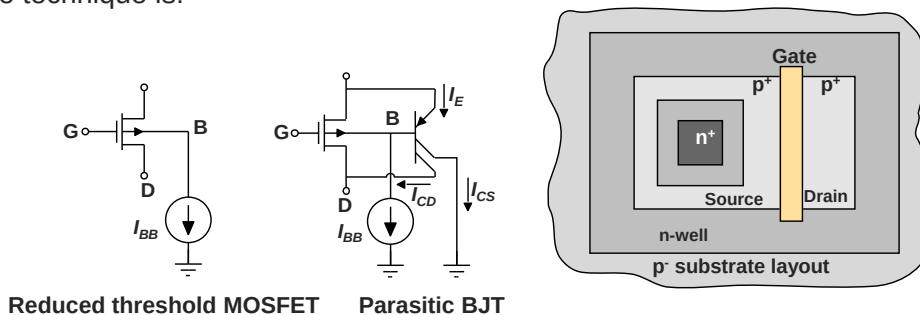


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Current-Driven Bulk Technique

The bulk can be used to reduce the threshold sufficiently to permit low-voltage applications. The key is to keep the substrate current confined.

One possible technique is:



Note: Need to limit the BJT current to some value called $I_{\max}$. Therefore,
$$I_{BB} = \frac{I_{\max}}{\beta_{CS} + \beta_{CD} + 1}$$

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Current-Driven Bulk Technique (continued)

Bias circuit for keeping the I_{max} defined independent of BJT betas.

Notice that:

$$I_{D,C} = I_{CD} + I_D$$

$$I_{S,E} = I_D + I_E + I_R$$

The circuit feedback causes a bulk bias current I_{BB} and hence a bias voltage V_{BIAS} such that:

$$I_{S,E} = I_D + I_{BB}(1 + \beta_{CS} + \beta_{CD}) + I_R$$

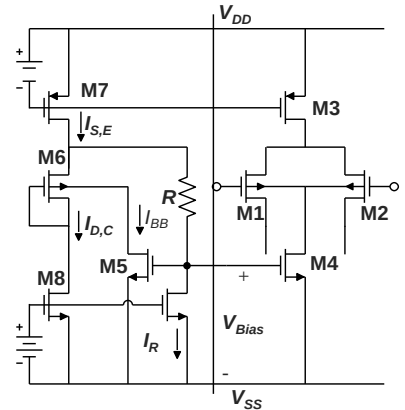
regardless of the actual values of the β 's.

Use V_{Bias1} and V_{Bias2} to set $I_{D,C} \approx 1.1I_D$, $I_{S,E} \approx 1.3I_D$ and $I_R \approx 0.1I_D$ which sets I_{max} at $0.1I_D$.

For the circuit to work:

$$V_{BE} < V_{TN} + I_R R \quad \text{and} \quad |V_{TP}| + V_{DS}(sat) < V_{TN} + I_R R$$

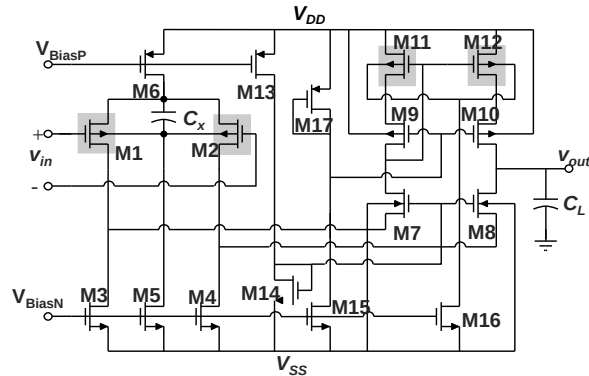
If $|V_{TP}| > V_{TN}$, then the level shifter I_{RR} can be eliminated.



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1V Folded-Cascode OTA: Current-Driven Bulk Technique

For large common-mode input changes, C_x is necessary to avoid slewing in the input stage. To get more voltage headroom at the output, the transistors of the cascode mirror have their bulks current driven.



A 1V folded-cascode operational transconductance amplifier (OTA) using the current-driven bulk technique

Note: Transistors with forward-biased bulks are in shaded boxes.

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Current-Driven Bulk Technique (continued)

Experimental results: 0.5 μm CMOS, 40 μA total bias current ($C_x = 10$ pF)

Supply voltage	1.0V	0.8V	0.7V
Common-mode input range	0.0V-0.65V	0.0V-0.4V	0.0V-0.3V
High-gain output range	0.35V-0.75V	0.25V-0.5V	0.2V-0.4V
Output saturation limits	0.1V-0.9V	0.15V-0.65V	0.1V-0.6V
DC gain	62 dB-69 dB	46 dB-53 dB	33 dB-36 dB
Gain-bandwidth	2.0 MHz	0.8 MHz	1.3 MHz
Slew rate ($C_L=20$ pF)	0.5 V/ μs	0.4 V/ μs	0.1 V/ μs
Phase margin ($C_L=20$ pF)	57°	54°	48°

The nominal value of bulk current is 10 nA gives a 10% increase in differential pair quiescent current assuming a BJT β of 100.



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Summary: Opamp Topologies

There are many techniques for designed amplifiers that meet the following requirements:

- As IC geometries decrease, the need for low-voltage designs becomes more critical.
- As the overall semiconductor area increases, power dissipation becomes a major design factor.
- For driving off-chip, low output resistances are often needed.



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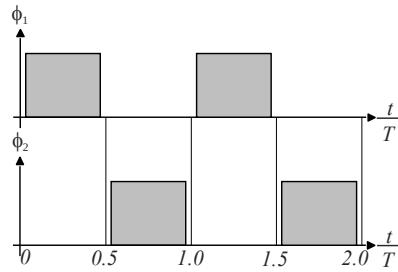
Introduction: Switched-Capacitor Amplifiers

- Switched-capacitor (switched-cap) amplifiers are time-varying circuits that yield circuits with smaller parasitic capacitors and therefore higher frequency response. Such circuits are *dynamically biased*.
- Switched-cap amplifiers require a nonoverlapping clock.
- Switched-cap amplifiers work only during a portion of a clock period.
- Bias conditions are set up on one clock phase and then maintained by capacitance on the active^{*} phase.
- Switched-cap amplifiers use switches and capacitors resulting in feedthrough problems.
- Simplified circuits on the active phase minimize the parasitics.



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Typical Switched Amplifier Clock

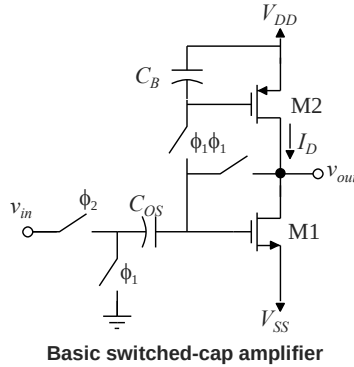


Nonoverlapping clock

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Dynamically Biased Inverting Amplifier

- During phase 1, the offset and bias of the inverter is sampled and applied to C_{OS} and C_B .
- During phase 2 C_{OS} is connected in series with the input and provides offset canceling plus bias for M1. C_B provides the bias for M2.



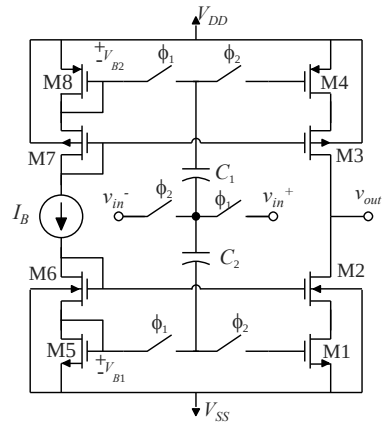
Note: This circuit illustrates the concept of switched amplifiers but is too simple to illustrate the reduction of bias parasitics.

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Dynamically Biased Push-Pull Cascode Amplifier

- **Push-pull, cascode amplifier:** M1-M2 and M3-M4.
- **Bias circuitry:** M5-M6-C2 and M7-M8-C1.

Parasitics can be further reduced by using a double-poly process to eliminate bulk-drain and bulk-source capacitances at the drain of M1-source of M2 and drain of M4-source of M3.

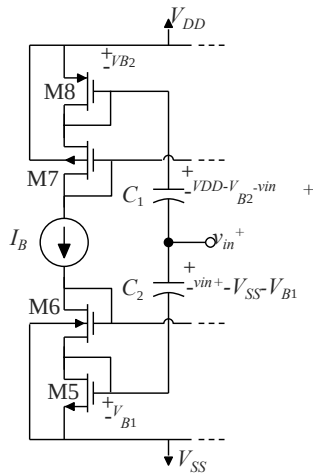


Switched-cap push-pull cascode amplifier

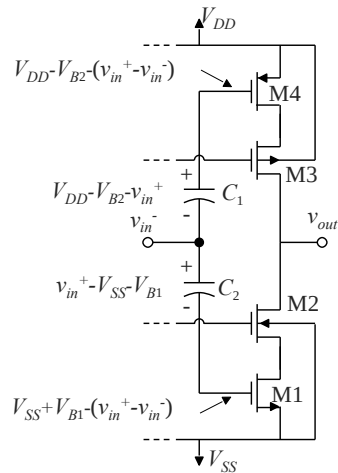


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Dynamically Biased (continued)



Equivalent circuit during the ϕ_1 clock period



Equivalent circuit during the ϕ_2 clock period

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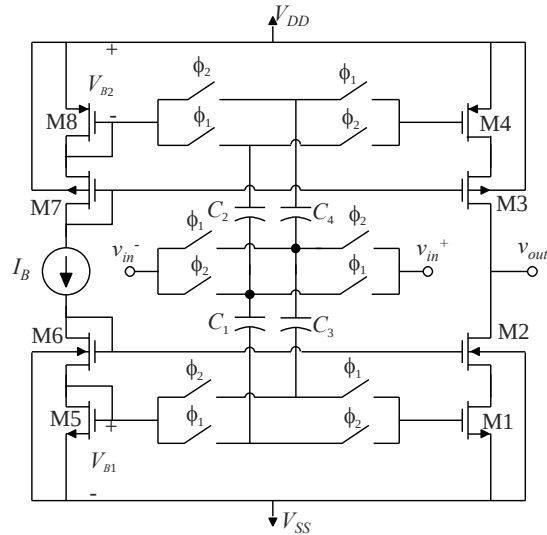
Dynamically Biased (continued)

This circuit operates on both clock phases*:

Performance (1.5 μm CMOS)

- 1.6 mW dissipation
- $GB \approx 130 \text{ MHz}$ ($C_L = 2.2 \text{ pF}$)
- Settling time of 10 ns ($C_L = 10 \text{ pF}$)

This amplifier was used with a 28.6 MHz clock to realize a fifth-order switched-capacitor filter having a cutoff frequency of 3.5 MHz.



Switched-cap push-pull cascode amplifier using both clock phases

*S. Masuda, et. al., "CMOS Sampled Differential Push-Pull Cascode Op Amp," *Proceedings of 1984 International Symposium on Circuits and Systems*, Montreal, Canada, May 1984, pp. 1211-1214.

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Summary: Switched-Capacitor Amplifiers

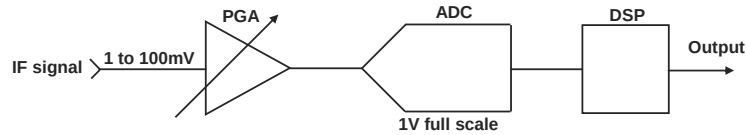
- Switched-cap amplifiers are time-varying circuits that yield circuits with higher frequency response.
- These amplifiers are dynamically biased.
- Switched-cap amplifiers require a nonoverlapping clock for proper operation.



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Introduction: Programmable Gain Amplifiers

- Programmable gain amplifiers (PGAs) are amplifiers with additional inputs that control their gains.
- PGAs are used in situations where signals of variable levels need to be amplified to fixed levels. For example, radio receivers commonly require PGAs. The received signal can vary over a large range but the radio demodulator (or an ADC) requires a signal of a certain amplitude.



Typical use of PGA

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Types of PGAs

- **Analog control:** Gain is proportional to an analog voltage or current.
- **Digital control:** One of a discrete number of gain values is selected by a digital control word.
- Analog-digital comparison
 - Analog control has the advantage of inherently fine gain control. With digital control, fine gain steps require a large number of control bits and a more complex amplifier.
 - Analog control has the advantage that the gain can be changed without introducing a discontinuity in the signal.
 - Digital control has the advantage of consistent performance regardless of process corner.
 - Analog control is susceptible to noise coupling in from the gain control path.
 - PGAs with analog control are generally harder to optimize for best noise performance and linearity.



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PGAs with Analog Control

In analog amplifiers, the gain usually depends on transconductances, or the gain is set by feedback network.

Changing g_m

- In bipolar $g_m = I_c/V_T$, so I_c can be easily used to implement gain control.
- Also possible in CMOS, but with less accuracy.
- Usually also noise, linear range, and output-drive capability changes.

Changing feedback

- Usually, MOS transistors are used as variable resistors. However, MOS resistors are not very linear.
- A change in feedback factor changes the loop gain and therefore closed-loop bandwidth, that is, bandwidth becomes usually lower at high-gain settings.

Some circuits combine both methods

- Active feedback circuits
- Extending gain-setting range



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PGAs with Digital Control

Digital control is usually relatively easy to implement, especially in CMOS:

- Design a fixed-gain amplifier to the required specifications.
- Introduce transmission-gate switches at the points where gain is set:
 - Switch opamp feedback resistors
 - Change bias currents in differential pair or operational transconductance amplifiers (OTAs).
 - Use amplifiers in parallel and switch between them.



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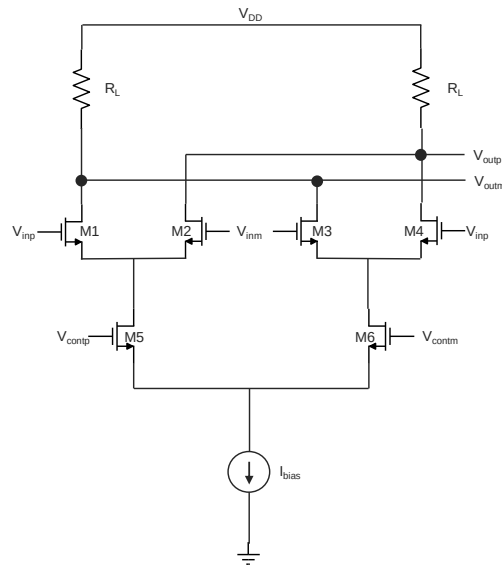
Example: Analog Control

$$g_{m1} = g_{m2} = g_{m3} = g_{m4} = g_{ms}$$

$$g_{m5} = g_{m6} = g_{mc}$$

$$A_v = \frac{(v_{outp} - v_{outm})}{(v_{inp} - v_{inm})}$$

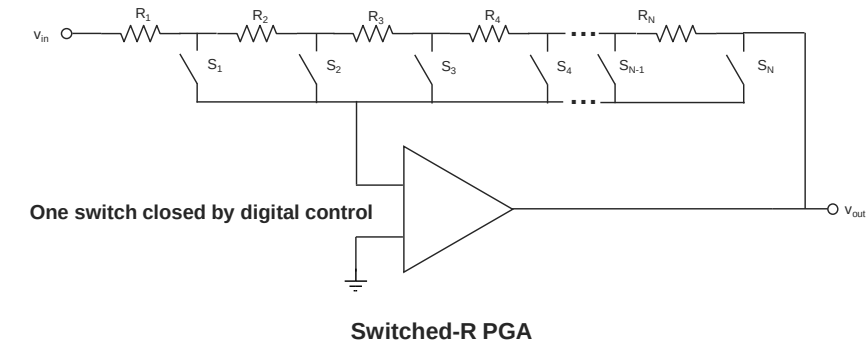
$$A_v \propto (v_{contp} - v_{contm})$$



PGA with analog control

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Diagram: Switched-R PGA



$$A_v = \frac{\sum_{i=m+1}^N R_i}{\sum_{i=1}^m R_i} \text{ (for switch } m \text{ closed)}$$

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Example: Switched-R PGA

Problem

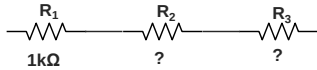
Select resistor values for the digital PGA example topology to provide gain of 0, 10, or 20 dB starting with $R_1=1\text{ K}\Omega$.

Solution

$$A_{v0dB} = 1$$

$$A_{v10dB} = 3.16$$

$$A_{v20dB} = 10$$



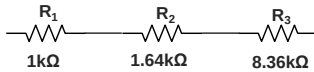
$$\frac{R_2 + R_3}{1k} = 10 \Rightarrow R_2 + R_3 = 10k$$

$$\frac{R_3}{1k + R_2} = 3.16 \Rightarrow R_3 = 3.16k + 3.16R_2$$

$$R_3 = 3.16k + 3.16(10k - R_3)$$

$$R_3 = 34.78k - 3.16R_3$$

$$R_3 = 8.36k \text{ \& \therefore } R_2 = 1.64k$$



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Typical PGA Problems

Transistors in amplifiers have a certain W and L, but changing feedback or bias currents leads to more compromises, especially if the gain range is large.

Not only gain might change, but also

- Noise level
- Input signal range and linearity
- Bandwidth

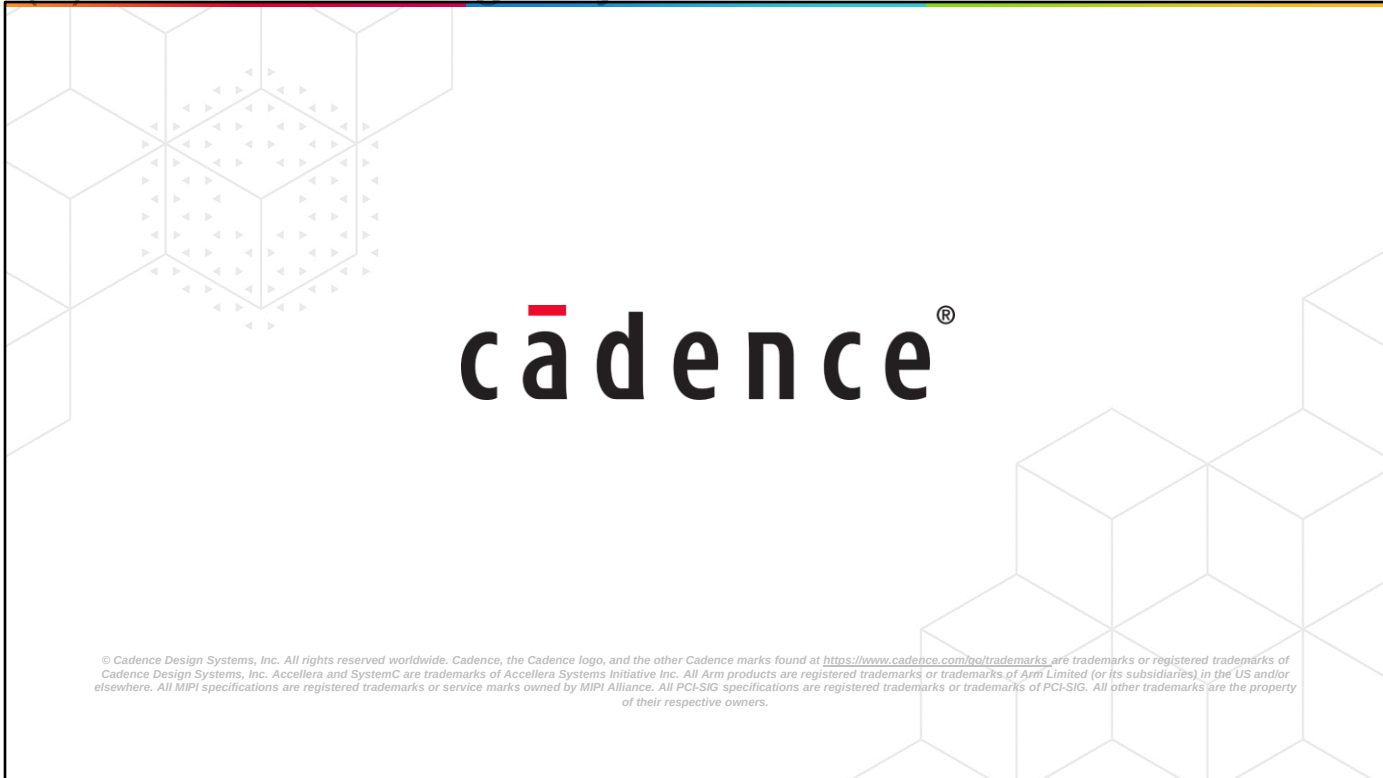
Circuitry to implement gain control is nonideal:

- CMOS transmission gates have series resistances and capacitances.
- For continuous control, often a linear or linear-in-dB gain setting characteristic is desirable, but it is not so easy to achieve.

Often you want from an amplifier multiple things such as treating a wide range of common-mode signals, doing a balanced-to-unbalanced conversion, or filtering. This might be hard to combine with gain control.



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